

OPTIMAL FEEDBACK LAW RECOVERY BY GRADIENT-AUGMENTED NEURAL NETWORK WITH NONCONVEX REGULARIZATION ON THE UNBOUNDED PARAMETER SPACE

ABSTRACT. We recover optimal feedback laws for nonlinear control problems by approximating Hamilton–Jacobi–Bellman value functions with sparse shallow neural networks. Value and gradient data are generated from the open-loop problem by the Pontryagin maximum principle, and the network is trained with a gradient-augmented H^1 fidelity. We adopt the measure representation of the network on the unbounded parameter space $\Omega = \mathbb{R}^{d+1}$. We use nonconvex regularization to enhance sparsity and treat the unbounded dictionaries in two cases. For nonhomogeneous activation functions of polynomial growth, including tanh, softplus, and the Gaussian, we introduce the weighted measure norm

$$\|\mu\|_{\mathcal{M}_p(\Omega)} = \int_{\Omega} (1 + |\omega|^p) d|\mu|(\omega).$$

Existence follows from the direct method of the calculus of variations using narrow convergence. For positively k -homogeneous activations, we derive an equivalent formulation on the unit sphere with the induced fractional-power penalty on the outer coefficients. In both formulations, we prove existence of global minimizers represented by finite networks and derive the corresponding first-order optimality conditions. In the homogeneous case, a positive lower bound on the nonzero outer coefficients gives an explicit upper bound on the network width. The two resulting primal–dual active point (PDAP) algorithms are studied on the Van der Pol oscillator and the pendulum swing-up problem, whose value function has a gradient discontinuity across a switching set.

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1. INTRODUCTION

The problem under consideration is the dynamic optimal control problem in the space-time domain

$$(P) \quad \begin{aligned} & \inf_{u \in L^2(t_0, T; \mathbb{R}^{m_u})} \mathcal{C}(u; t_0, x) := \int_{t_0}^T (h(y(t)) + \eta |u(t)|^2) dt, \quad \eta > 0, \\ & \text{s.t.} \\ & \dot{y} = f(y) + g(y)u, \\ & y(t_0) = x, \end{aligned}$$

We assume $y(t) \in \mathbb{R}^d$, $u(t) \in \mathbb{R}^{m_u}$, $h \in C^1(\mathbb{R}^d)$, $f \in C^1(\mathbb{R}^d; \mathbb{R}^d)$, and $g \in C^1(\mathbb{R}^d; \mathbb{R}^{d \times m_u})$. We assume throughout that a minimizer of (P) exists.

One approach computes an open-loop state-costate-control path for a fixed initial condition x . The Pontryagin maximum principle (PMP) yields a coupled forward-backward two-point boundary value problem:

$$(TPBVP) \quad \begin{cases} \frac{d}{dt} y^* = f(y^*) + g(y^*)u^*, \\ y^*(t_0) = x, \\ -\frac{d}{dt} p^* = \partial_y \left(f(y^*) + g(y^*)u^* \right)^\top p^* + \partial_y h(y^*), \\ p^*(T) = 0. \end{cases}$$

The stationarity condition with respect to the control gives

$$(1) \quad u^* = -\frac{1}{2\eta} g(y^*)^\top p^* \quad \forall t \in (t_0, T).$$

The open-loop control is not robust with respect to disturbances of the state [1]: it is computed for a single initial condition, and deviations from the optimal path are not corrected. In addition, numerical methods for the two-point boundary value problem are sensitive to the initial guess and may fail to converge. The dynamic programming principle instead introduces the value function

$$V(t, x) := \inf_u \mathcal{C}(u; t, x)$$

which satisfies the Hamilton–Jacobi–Bellman (HJB) equation:

$$(HJB) \quad \begin{cases} \partial_t V - \frac{1}{4\eta} \nabla V(t, x)^\top g(x) g(x)^\top \nabla V(t, x) + \nabla V(t, x)^\top f(x) + h(x) = 0, \\ V(T, x) = 0. \end{cases}$$

The value function encodes the optimal cost from every state. The feedback law synthesized from it corrects deviations from a nominal path.

For nonlinear control problems one usually seeks a feedback law, which requires solving the HJB equation. Computing this function is notoriously difficult because of the curse of dimensionality. Alongside classical methods, data-driven methods have therefore become increasingly common. One line of work, introduced in [21], generates the training data with (TPBVP) and approximates the optimal value function from it. The optimal feedback law is then synthesized from the fitted value function after validation. Its samples are causality-free: generating one does not require approximate values at neighbouring grid points. Moreover, Azmi, Kalise and Kunisch [1] enrich the dataset with the gradient of the value function: a state-adjoint-control triple is generated by

solving the open-loop problem, and the adjoint supplies the gradient. The value function is then approximated by hyperbolic-cross basis functions with an ℓ^1 penalty, and including the adjoint data targets an H^1 approximation. Neural-network approximations form another growing line of work [19, 28].

For $\omega = (\omega_n)_{n=1}^N$, $\omega_n = (a_n, b_n)$, and $c = (c_n)_{n=1}^N$, we use the two-layer network

$$(2) \quad \mathcal{N}_{\omega,c}(x) := \sum_{n=1}^N c_n \rho(a_n \cdot x + b_n) \in \mathbb{R},$$

where ρ is a sufficiently smooth activation function and N the width of the network. We associate it with the finite signed measure $\mu_{\omega,c} := \sum_{n=1}^N c_n \delta_{\omega_n}$. Its gradient is

$$\nabla_x \mathcal{N}_{\omega,c}(x) = \sum_{n=1}^N c_n \nabla_x \rho(a_n \cdot x + b_n) \in \mathbb{R}^d.$$

The training data are generated by solving (TPBVP). The adjoint state p^* is obtained as a byproduct and provides the gradient information of the optimal value function: $p^*(t) = \nabla V(t, y^*(t))$ wherever $V(t, \cdot)$ is differentiable at $y^*(t)$ [10, 8, 9]. Where V is not differentiable — on the switching sets studied in Section 6 — competing optimal paths may have distinct costates, and each sampled p^* gives the corresponding one-sided gradient of V . Thus the data consist of triples $\{x^i, V(x^i), p^{*,i}\}_{i=1}^M$; wherever V is differentiable, $p^{*,i} = \nabla V(x^i)$. For the finite-horizon problem, we write $V(x) := V(0, x)$. The infinite-horizon pendulum value considered below is stationary, so it has no suppressed time argument. We have the minimization problem

$$(3) \quad \min_{N, \omega, c} l^M(\mu_{\omega,c}), \quad l^M(\mu_{\omega,c}) := \frac{1}{2M} \sum_{m=1}^M (|V(x^m) - \mathcal{N}_{\omega,c}(x^m)|^2 + |\nabla V(x^m) - \nabla \mathcal{N}_{\omega,c}(x^m)|^2).$$

Section 3 introduces the corresponding population fidelity L with respect to Lebesgue measure on D .

A traditional sparse shallow network uses the ReLU activation and an ℓ^1 penalty on the outer coefficients [5, 2]. Under gradient-augmented fitting, this formulation has two limitations. 1. The convex penalty may retain redundant neurons whose inner parameters cluster. 2. The second-order weak derivative of ReLU does not exist as a function while it is a distribution, and we cannot use it to approximate high-order derivatives.

Sparsity beyond the convex penalty. The number of neurons N is itself an optimization variable. Although the number of neurons does not act as the inductive bias of the model in general [22], sparsity is crucial for computational efficiency in many problems. Furthermore, sparsity is necessary for certain applications, such as knowledge distillation and deconvolution of point-like objects. A standard sparse-learning procedure uses a convex reformulation with a sparsity-promoting penalty on the outer weights, most commonly the ℓ^1 penalty (LASSO). Pieper and Petrosyan [25] showed that convex penalties cannot effectively eliminate this redundancy when neurons cluster in a small neighborhood and introduced a concave nonconvex functional for this purpose. We extend their nonconvex measure formulation beyond the ReLU activation function for the gradient-augmented fitting on an unbounded inner-parameter space using the concave functional. Its strict local concavity yields finite support for the nonhomogeneous formulation, while the induced fractional exponent gives a quantitative width bound for positively k -homogeneous activations.

Activation functions beyond ReLU. We therefore consider smoother activation functions. This extension requires a treatment of the unbounded inner-parameter space for gradient training. For positively k -homogeneous activations, radial scaling can be absorbed into the outer coefficient, and the network has an equivalent representation with inner parameters on the unit sphere. For nonhomogeneous activations such as tanh, softplus, and the Gaussian, radial scaling changes

the atom and cannot be removed by normalization. Without additional control, sequences with $|\omega| \rightarrow \infty$ may retain a nonzero contribution to the represented function, as shown in Section 3.3. We therefore introduce

$$\|\mu\|_{\mathcal{M}_p(\Omega)} := \int_{\Omega} (1 + |\omega|^p) d|\mu|(\omega).$$

This weighted total-variation norm gives tightness and provides the compactness used in the existence proof. We analyze the resulting variational problem $J(\mu) = L(\mu) + \alpha\Phi_{\phi}(\mu) + \beta\|\mu\|_{\mathcal{M}_p(\Omega)}$ over measures on the unbounded parameter domain.

Contributions. The contributions are the following.

- (1) *Nonhomogeneous activations on the unbounded parameter domain.* For continuous activation functions of polynomial growth $|\rho(z)| \leq C_{\rho}(1 + |z|)^{s_0}$ and moment order $p > s_0$, we prove existence of global minimizers of the gradient-augmented, moment-regularized problem over measures on $\Omega = \mathbb{R}^{d+1}$ (Theorem 3.16). The class covered includes tanh, the Gaussian ($s_0 = 0$) and softplus ($s_0 = 1$), for which the $\beta = 0$ problem loses compactness or becomes nonattaining (Examples 3.10 and 3.11). The Gâteaux derivative obeys $|\bar{P}(\omega)| \leq C_K \|r_{\bar{\mu}}\|_{H^1(D)}(1 + |\omega|)^{s_1}$, while the moment weight grows like $|\omega|^p$. When $p > s_1$, every support point lies in an explicit ball (Lemma 3.21). Local minimizers are therefore finitely supported (Theorem 3.23), and the strict optimality inequality only needs to be checked on a bounded set (Theorem 3.28). The first-order conditions contain the moment weight in the pointwise inequality $|\bar{P}(\omega)| \leq \alpha + \beta w_p(\omega)$ (Theorem 3.20), and the same inequality gives the greedy acceptance rule in Section 5.
- (2) *The k -homogeneous formulation.* For positively k -homogeneous activations, we derive an equivalent problem on the unit sphere with the fractional-power penalty $|c|^{2/(k+1)}$ (Propositions 4.3 and 4.5). We also derive the first-order optimality conditions and prove existence of a global minimizer with an explicit upper bound on the network width (Corollaries 4.6 and 4.8).
- (3) *Gradient-augmented algorithms and feedback recovery.* The first insertion algorithm combines the log penalty with the moment norm; the second uses the fractional-exponent penalty on the unit sphere. The numerical examples compare activation regularity, support size, and feedback recovery for a smooth value function and for a value function whose gradient is discontinuous across a switching set.

Figure 1 compares the traditional ReLU network with an ℓ^1 penalty against representative models from the two formulations. The Gaussian and ReLU² reduce the relative H^1 error on the Van der Pol problem, and ReLU² gives the lowest error on the pendulum problem.

The thesis is organized as follows. Section 2 summarizes the general measure-theoretic and approximation results needed for the proofs. Section 3 introduces the learning problem over measures, exhibits the failure of the $\beta = 0$ problem on the unbounded parameter domain, and develops the existence, optimality, and finite-support theory for the nonhomogeneous activations. Section 4 treats the k -homogeneous reformulation and the fractional-exponent penalty. Section 5 describes the data generation and the two insertion algorithms, and Section 6 reports the numerical studies.

The overall pipeline has four steps:

- (1) **data generation:** initially, a dataset of causality-free state-value-gradient triples must be generated. Many methods are discussed in the literature [15, 12]. The process is highly parallelizable and adaptive: dense samples can be placed in regions of interest.
- (2) **model selection:** depending on the underlying system, different activation functions and penalties can be chosen. The activation function and the regularization affect the approximation error and support size. We can also incorporate other prior information such as semiconcavity of the value function or the physics-informed loss [11, 20, 26, 29, 18].

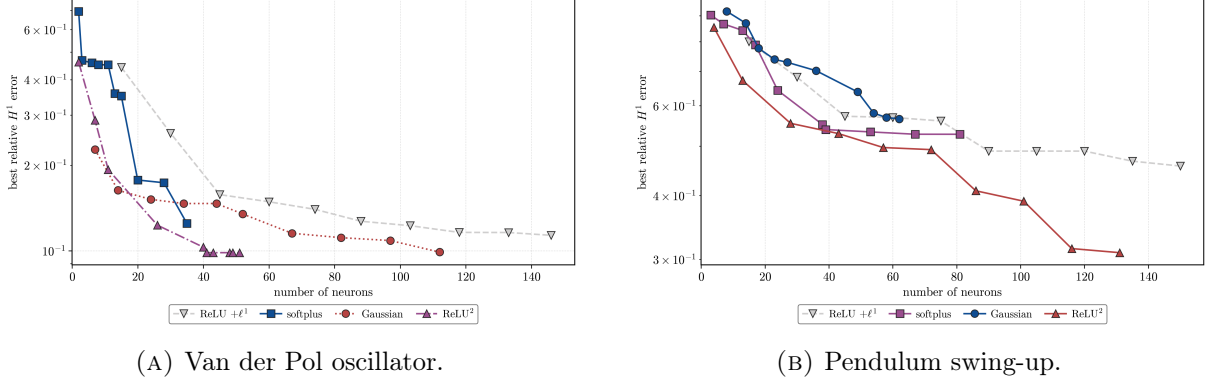


FIGURE 1. Running best relative H^1 validation error as neurons are inserted. Van der Pol parameters: ReLU with ℓ^1 , $\alpha = 10^{-5}$; softplus, $(\alpha, \beta, p, \gamma) = (10^{-5}, 10^{-10}, 2.01, 1)$; Gaussian, $(10^{-5}, 10^{-10}, 3, 1)$; and ReLU², $\alpha = 10^{-5}$. Pendulum parameters: ReLU with ℓ^1 , $\alpha = 10^{-5}$; softplus, $(10^{-4}, 10^{-10}, 2.01, 1)$; Gaussian, $(10^{-4}, 10^{-4}, 2.01, 0)$; and ReLU², $\alpha = 10^{-6}$. All curves use seed 42.

- (3) **training and validation:** this step follows the standard machine-learning pipeline. A neural network is trained to approximate the optimal value function, and the performance of the model is subsequently evaluated on a separate validation set. The validation set can be chosen to probe specific regions, such as the switching set.
- (4) **feedback control:** once the value function is trained, the feedback law follows from the Hamiltonian minimization. Sparsity keeps the evaluation of the law cheap.

2. PRELIMINARIES

This section reviews general theories used in the thesis. The learning objectives and problem-specific assumptions are introduced in Section 3.

2.1. Finite signed Radon measures. Let Ω be a locally compact Polish space. We write $\mathcal{M}(\Omega)$ for the Banach space of finite signed Radon measures on Ω , equipped with the total-variation norm

$$\|\mu\|_{\mathcal{M}(\Omega)} := |\mu|(\Omega).$$

By the Riesz representation theorem, $\mathcal{M}(\Omega) = C_0(\Omega)^*$ under the pairing

$$\langle f, \mu \rangle := \int_{\Omega} f d\mu.$$

Here $C_0(\Omega)$ denotes the continuous functions vanishing at infinity and $C_b(\Omega)$ the bounded continuous functions.

Let $\mathcal{M}^0(\Omega)$ and $\mathcal{M}^\sharp(\Omega)$ denote, respectively, the nonatomic and purely atomic measures.

For $\mu \in \mathcal{M}(\Omega)$ we define the *atom set* $\text{atom } \mu := \{\omega \in \Omega : |\mu|(\{\omega\}) > 0\}$, which is at most countable, the purely atomic part $\mu_{\text{atom}} := \sum_{\omega \in \text{atom } \mu} \mu(\{\omega\}) \delta_{\omega} \in \mathcal{M}^\sharp(\Omega)$, and the nonatomic part $\mu_{\text{cont}} := \mu - \mu_{\text{atom}} \in \mathcal{M}^0(\Omega)$ [7]. This gives the decomposition

$$\mathcal{M}(\Omega) = \mathcal{M}^0(\Omega) \oplus \mathcal{M}^\sharp(\Omega).$$

The atom set need not be closed because atoms may accumulate at a point that carries no mass. The polar decomposition of a signed measure is written as

$$\mu = \text{sign}(\mu) |\mu|, \quad \text{sign}(\mu) := \frac{d\mu}{d|\mu|} \in \{-1, 1\} \quad |\mu|\text{-almost everywhere.}$$

2.2. Narrow convergence, tightness, and Prokhorov's theorem.

Definition 2.1 (Narrow convergence for signed measures). For $\mu_n, \mu \in \mathcal{M}(\Omega)$ we write $\mu_n \rightarrow_n \mu$ if

$$\int_{\Omega} f d\mu_n \rightarrow \int_{\Omega} f d\mu \quad \text{for every } f \in C_b(\Omega).$$

Since $C_0(\Omega) \subset C_b(\Omega)$, narrow convergence implies weak-* convergence. The converse need not hold on a noncompact domain.

Definition 2.2 (Uniform tightness). A family $\mathcal{A} \subset \mathcal{M}(\Omega)$ is uniformly tight if, for every $\varepsilon > 0$, there is a compact set $K_\varepsilon \subset \Omega$ such that

$$\sup_{\mu \in \mathcal{A}} |\mu|(\Omega \setminus K_\varepsilon) < \varepsilon.$$

Theorem 2.3 (Prokhorov's theorem, [6]). *Let $\mathcal{A}$ be a family of nonnegative finite Radon measures on Ω . Every sequence in $\mathcal{A}$ has a narrowly convergent subsequence if and only if $\mathcal{A}$ is uniformly tight and $\sup_{\mu \in \mathcal{A}} \mu(\Omega) < \infty$.*

Corollary 2.4. *Let $(\mu_n) \subset \mathcal{M}(\Omega)$ satisfy*

$$\sup_n |\mu_n|(\Omega) < \infty$$

and suppose that the family $(|\mu_n|)$ is uniformly tight. Then (μ_n) has a narrowly convergent subsequence.

Proof. Write the Jordan decomposition $\mu_n = \mu_n^+ - \mu_n^-$. Since $\mu_n^\pm \leq |\mu_n|$, both nonnegative families are uniformly bounded in mass and uniformly tight. Apply Prokhorov's theorem first to (μ_n^+) and then to (μ_n^-) along the resulting subsequence. Thus $\mu_n^+ \rightarrow_n \bar{\mu}^+$ and $\mu_n^- \rightarrow_n \bar{\mu}^-$ along a common subsequence, and consequently $\mu_n \rightarrow_n \bar{\mu}^+ - \bar{\mu}^-$. $\square$

2.3. Measures with a finite moment. In this subsection let $\Omega = \mathbb{R}^{d_\Omega}$, where $d_\Omega \geq 1$. For $p > 0$, set

$$w_p(\omega) := 1 + |\omega|^p$$

and define

$$(4) \quad \mathcal{M}_p(\Omega) := \left\{ \mu \in \mathcal{M}(\Omega) : \int_{\Omega} w_p d|\mu| < \infty \right\}, \quad \|\mu\|_{\mathcal{M}_p(\Omega)} := \int_{\Omega} w_p d|\mu|.$$

The associated space of continuous functions is

$$C_{w_p^{-1}}(\Omega) := \left\{ f \in C(\Omega) : \frac{f}{w_p} \in C_0(\Omega) \right\}, \quad \|f\|_{C_{w_p^{-1}}} := \sup_{\omega \in \Omega} \frac{|f(\omega)|}{w_p(\omega)}.$$

Under the dual pairing, $\mathcal{M}_p(\Omega) = C_{w_p^{-1}}(\Omega)^*$ isometrically [4, Section 5]. For an atomic measure,

$$(5) \quad \left\| \sum_{j=1}^N c_j \delta_{\omega_j} \right\|_{\mathcal{M}_p(\Omega)} = \sum_{j=1}^N |c_j| (1 + |\omega_j|^p).$$

2.4. Barron spaces. In view of the approximation by neural networks, the seminal paper [3] develops a universal approximation theorem on function approximation by feedforward neural networks with sigmoidal activation functions. Later, due to the popularity of ReLU, the approximation properties of networks with ReLU activation are investigated in the subsequent studies [23, 27, 5, 24]. In particular, it is argued that generalization is achieved not by limiting the size of the network but by controlling the magnitude of the weights. Let $\mathcal{W}(D)$ be the space of functions f on D that can be represented by $f(x) = \int_{\mathbb{S}^d} (a \cdot x + b)_+ d\mu(a, b)$ for every $x \in D$ and some $\mu \in \mathcal{M}(\mathbb{S}^d)$. For $f \in \mathcal{W}(D)$, we define the norm

$$(6) \quad \|f\|_{\mathcal{W}(D)} = \inf \left\{ \|\mu\|_{\mathcal{M}(\mathbb{S}^d)} : f(x) = \int_{\mathbb{S}^d} (a \cdot x + b)_+ d\mu(a, b) \text{ for every } x \in D \right\}.$$

It is shown that this norm is closely related to a fractional Laplacian of f [23, 27].

For ReLU^k with $k \geq 2$, we follow the extended Barron spaces of [17]. There, a function of the form

$$f(x) = \int c(a \cdot x + b)_+^k d\varrho(a, b, c),$$

with ϱ a *probability* measure over the parameters $(a, b) \in \mathbb{R}^{d+1}$ and the outer weight $c \in \mathbb{R}$, is assigned the representation cost

$$\|f\|_{B_{1,\varrho}^k} := \int |c| (\|a\|_1 + |b|)^k d\varrho(a, b, c),$$

and the Barron norm is the infimum over all representing probabilities,

$$\|f\|_{B_1^k} := \inf_{\varrho} \|f\|_{B_{1,\varrho}^k}.$$

The space $B_1^k(D)$ of functions on D with finite Barron norm is a normed vector space [17, Theorem 6].

Proposition 2.5 (Barron–Sobolev embedding, [17, Theorem 7]). *Given any fixed integer k , there holds $B_1^k(D) \subset H^k(D)$; moreover there is a constant $C(D, d, k)$ such that*

$$\|f\|_{H^m(D)} \leq C(D, d, k) \|f\|_{B_1^k}, \quad 0 \leq m \leq k,$$

for all $f \in B_1^k(D)$.

For a Lipschitz activation ρ , let

$$\mathcal{G}_{\rho,f} := \left\{ \mu \in \mathcal{M}(\mathbb{R}^{d+1}) : f(x) = \int_{\mathbb{R}^{d+1}} \rho(a \cdot x + b) d\mu(a, b) \text{ for every } x \in D \right\}.$$

Following [13], define

$$\|f\|_{B_\rho} := \inf_{\mu \in \mathcal{G}_{\rho,f}} \int_{\mathbb{R}^{d+1}} (1 + \|a\|_1 + |b|) d|\mu|(a, b), \quad B_\rho := \{f : \|f\|_{B_\rho} < \infty\}.$$

Proposition 2.6 (Embeddings into ReLU^k Barron spaces, [13, Theorem 1]). *Let ρ be a Lipschitz activation function and let D be bounded.*

- (1) *If $\rho \in C^1(\mathbb{R})$ and $\partial^2 \rho \in L^1(\mathbb{R})$, then $B_\rho \hookrightarrow B_{\text{ReLU}}$.*
- (2) *If $k \geq 1$, $\rho \in C^k(\mathbb{R})$, and the one-sided (Caputo) derivatives of order $k+1$ satisfy $\partial_+^{k+1} \rho \in L^1((0, \infty))$ and $\partial_-^{k+1} \rho \in L^1((-\infty, 0))$ — for which it suffices that the $(k+1)$ -st distributional derivative of ρ exists as a finite measure — then*

$$B_\rho \hookrightarrow B_{\text{ReLU}^k}.$$

Notation. Throughout the remainder of the thesis, $D \subset \mathbb{R}^d$ is a bounded Lipschitz domain, $\nu := \mathcal{L}^d \llcorner D$ is the population measure, and $L^2(D)$ means $L^2(D, \nu)$. The symbol M denotes the number of samples in an empirical loss. The inner-parameter domain is $\Omega = \mathbb{R}^{d+1}$, with

$$\omega = (a, b), \quad [K(\omega)](x) := \rho(a \cdot x + b).$$

For a finite network, we write

$$\omega = (\omega_n)_{n=1}^N, \quad c = (c_n)_{n=1}^N, \quad \mu_{\omega,c} := \sum_{n=1}^N c_n \delta_{\omega_n}, \quad \mathcal{N}_{\omega,c} := \sum_{n=1}^N c_n K(\omega_n).$$

The symbol δ_ω denotes the Dirac measure at ω , while μ' denotes a general signed-measure direction. The symbol $\mathcal{H}^{d-1}$ denotes $(d-1)$ -dimensional Hausdorff measure. We reserve p for the moment order.

3. THE LEARNING PROBLEM

3.1. The value operator and the gradient-augmented loss. Whenever the integral exists, we define the *value operator* of a measure $\mu \in \mathcal{M}(\Omega)$ by

$$[\mathcal{N}\mu](x) := \int_{\Omega} [K(\omega)](x) d\mu(\omega),$$

where $\rho \in C(\mathbb{R})$. In particular, $\mathcal{N}\mu_{\omega,c} = \mathcal{N}_{\omega,c}$.

Assumption 3.1 (Polynomial growth of ρ). There are $C_\rho > 0$ and $s_0 \geq 0$ such that

$$(7) \quad |\rho(z)| \leq C_\rho(1 + |z|)^{s_0} \quad (z \in \mathbb{R}).$$

Since D is bounded, (7) gives a constant C_D with

$$(8) \quad \|K(\omega)\|_{L^2(D)} \leq C_D(1 + |\omega|)^{s_0}.$$

Thus $\mathcal{N}\mu$ is a well-defined $L^2(D)$ -valued Bochner integral if $\int_{\Omega} (1 + |\omega|)^{s_0} d|\mu|(\omega) < \infty$; in particular this holds on $\mathcal{M}_p(\Omega)$ when $p > s_0$. The map $K : \Omega \rightarrow L^2(D)$ is continuous: if $\omega_n \rightarrow \omega$, all pre-activations $a_n \cdot x + b_n$ remain in one compact interval for $x \in D$, and uniform continuity of ρ on that interval gives $K(\omega_n) \rightarrow K(\omega)$ uniformly on D .

The gradient of the operator is understood distributionally; whenever

$$(9) \quad \int_{\Omega} \|\nabla K(\omega)\|_{L^2(D)} d|\mu|(\omega) < \infty,$$

the map $\omega \mapsto \partial_{x_i} K(\omega)$ is Bochner integrable into $L^2(D)$. Fubini's theorem against test functions gives

$$\nabla(\mathcal{N}\mu)(x) = \int_{\Omega} \nabla K(\omega)(x) d\mu(\omega).$$

We write $\nabla \mathcal{N}\mu := \nabla(\mathcal{N}\mu)$ in this case and set $r_\mu := \mathcal{N}\mu - V$.

The fidelity term is the gradient-augmented loss with the distributional gradient convention,

$$L(\mu) := \begin{cases} \frac{1}{2} \|\mathcal{N}\mu - V\|_{L^2(D)}^2 + \frac{1}{2} \|\nabla(\mathcal{N}\mu) - \nabla V\|_{L^2(D)}^2, & \mathcal{N}\mu \in H^1(D), \\ +\infty, & \text{otherwise.} \end{cases}$$

For sample points $x^1, \dots, x^M \in D$ and a finite network for which the displayed point values exist, the corresponding empirical fidelity is

$$(10) \quad l^M(\mu) := \frac{1}{2M} \sum_{m=1}^M (|r_\mu(x^m)|^2 + |\nabla r_\mu(x^m)|^2).$$

For value-only fitting, the gradient summand is omitted.

3.2. The concave nonconvex functional. Let $\phi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ satisfy the following assumptions.

Assumption 3.2. The function ϕ is differentiable at 0 and $\phi'(0) = 1$.

Assumption 3.3. The function ϕ is differentiable, concave, nondecreasing, and coercive, with $\phi(0) = 0$.

Lemma 3.4. If Assumption 3.3 holds, then

- (1) $\phi(z_1 + z_2) \leq \phi(z_1) + \phi(z_2)$ for all $z_1, z_2 \geq 0$;
- (2) if Assumption 3.2 also holds, then $\phi(z) \leq z$ for all $z \geq 0$.

Proof. Concavity makes ϕ' nonincreasing. Therefore

$$\begin{aligned} \phi(z_1 + z_2) &= \int_0^{z_1} \phi'(s) ds + \int_0^{z_2} \phi'(z_1 + s) ds \\ &\leq \int_0^{z_1} \phi'(s) ds + \int_0^{z_2} \phi'(s) ds = \phi(z_1) + \phi(z_2). \end{aligned}$$

If $\phi'(0) = 1$, then $\phi(z) = \int_0^z \phi'(s) ds \leq z$. $\square$

Assumption 3.5 (Local curvature). There are $\gamma_1 \geq 0$, $\gamma_2 > 0$, and $\hat{z}_1, \hat{z}_2 > 0$ such that

- (1) $-\gamma_1(z_2 - z_1) \leq \phi'(z_2) - \phi'(z_1) \leq 0$ for $0 \leq z_1 \leq z_2 \leq \hat{z}_1$;
- (2) $\phi'(z_2) - \phi'(z_1) \leq -\gamma_2(z_2 - z_1)$ for $0 \leq z_1 \leq z_2 \leq \hat{z}_2$.

Remark 3.6. Assumption 3.5 is local. Global concavity alone does not give the lower curvature bound in part (1). For example, $\phi(z) = z - \frac{2}{3}z^{3/2}$ near the origin has $\phi''(z) = -\frac{1}{2}z^{-1/2}$; it can be extended linearly after a small positive point so as to remain nondecreasing, concave, and coercive.

Proposition 3.7 (Second-order bounds). *Under Assumptions 3.2–3.5,*

- (1) $\phi(z) \geq z - \frac{\gamma_1}{2}z^2$ for $0 \leq z \leq \hat{z}_1$;
- (2) $\phi(z) \leq z - \frac{\gamma_2}{2}z^2$ for $0 \leq z \leq \hat{z}_2$.

Proof. Since $\phi(0) = 0$ and $\phi'(0) = 1$,

$$\phi(z) - z = \int_0^z [\phi'(s) - \phi'(0)] ds.$$

The two conclusions follow by integrating the corresponding bounds in Assumption 3.5. $\square$

Define

$$(11) \quad \Phi_\phi(\mu) := \|\mu_{\text{cont}}\|_{\mathcal{M}(\Omega)} + \sum_{\omega \in \text{atom } \mu} \phi(|\mu(\{\omega\})|).$$

The continuous part is weighted by $\phi'(0) = 1$. This is the weak-* lower-semicontinuous relaxation of the concave cost on the atoms [7].

Lemma 3.8 (Coercivity of Φ_ϕ). *Let ϕ satisfy Assumptions 3.2 and 3.3. Then, for every $\mu \in \mathcal{M}(\Omega)$,*

$$(12) \quad \phi(\|\mu\|_{\mathcal{M}(\Omega)}) \leq \Phi_\phi(\mu) \leq \|\mu\|_{\mathcal{M}(\Omega)}.$$

In particular, Φ_ϕ is coercive with respect to the total-variation norm.

Proof. The decomposition into the nonatomic and purely atomic parts gives

$$\|\mu\|_{\mathcal{M}(\Omega)} = |\mu|(\Omega \setminus \text{atom } \mu) + \sum_{\omega \in \text{atom } \mu} |\mu|(\{\omega\}).$$

The subadditivity in Lemma 3.4 extends to a countable sum: apply it first to finite partial sums and then use the continuity and monotonicity of ϕ . Hence

$$\begin{aligned} \phi(\|\mu\|_{\mathcal{M}(\Omega)}) &\leq \phi(|\mu|(\Omega \setminus \text{atom } \mu)) + \sum_{\omega \in \text{atom } \mu} \phi(|\mu|(\{\omega\})) \\ &\leq |\mu|(\Omega \setminus \text{atom } \mu) + \sum_{\omega \in \text{atom } \mu} \phi(|\mu|(\{\omega\})) = \Phi_\phi(\mu), \end{aligned}$$

where the second inequality uses $\phi(z) \leq z$. Applying the same bound to each atom gives

$$\Phi_\phi(\mu) \leq |\mu|(\Omega \setminus \text{atom } \mu) + \sum_{\omega \in \text{atom } \mu} |\mu|(\{\omega\}) = \|\mu\|_{\mathcal{M}(\Omega)}.$$

The first inequality in (12) and the coercivity of ϕ prove the final assertion. $\square$

Lemma 3.9 (Lower semicontinuity of Φ_ϕ). *Let ϕ satisfy Assumptions 3.2 and 3.3. If $\mu_n \rightharpoonup_n \mu$, then $\Phi_\phi(\mu) \leq \liminf_{n \rightarrow \infty} \Phi_\phi(\mu_n)$.*

Proof. Narrow convergence implies weak-* convergence against $C_0(\Omega)$. By the uniform boundedness principle,

$$M_0 := \sup_n \|\mu_n\|_{\mathcal{M}(\Omega)} < \infty.$$

After passing to a subsequence, it is enough to consider the case

$$\lim_{n \rightarrow \infty} \Phi_\phi(\mu_n) = \liminf_{n \rightarrow \infty} \Phi_\phi(\mu_n) =: \ell < \infty.$$

Fix $\varepsilon > 0$ and decompose each measure in the sequence as

$$\mu_n = \zeta_n + \eta_n, \quad \zeta_n := \mu_n|_{A_n}, \quad A_n := \{\omega \in \Omega : |\mu_n|(\{\omega\}) \geq \varepsilon\}.$$

Thus ζ_n contains the atoms whose masses are at least ε , whereas η_n contains the nonatomic part and all remaining atoms. Since the two measures have disjoint atom sets,

$$(13) \quad \Phi_\phi(\mu_n) = \Phi_\phi(\zeta_n) + \Phi_\phi(\eta_n).$$

The bound on the total variation shows that A_n contains at most $k := \lfloor M_0/\varepsilon \rfloor$ points. Moreover, $\|\zeta_n\|_{\mathcal{M}(\Omega)} \leq M_0$ and $\|\eta_n\|_{\mathcal{M}(\Omega)} \leq M_0$. Since $C_0(\Omega)$ is separable for $\Omega = \mathbb{R}^{d+1}$, weak-* compactness and a common subsequence give

$$\zeta_n \xrightarrow{*} \zeta, \quad \eta_n \xrightarrow{*} \eta, \quad \zeta + \eta = \mu.$$

Every ζ_n is purely atomic and has at most k atoms. By [7, Lemma 3.6], the set of such measures is sequentially weak-* closed, and the functional obtained by summing ϕ over these atoms is sequentially weak-* lower semicontinuous. Consequently,

$$(14) \quad \liminf_{n \rightarrow \infty} \Phi_\phi(\zeta_n) \geq \Phi_\phi(\zeta).$$

For $0 \leq z \leq \varepsilon$, concavity and $\phi(0) = 0$ give

$$\phi(z) \geq \frac{\phi(\varepsilon)}{\varepsilon} z =: (1 - \delta_\varepsilon)z,$$

where

$$\delta_\varepsilon := 1 - \frac{\phi(\varepsilon)}{\varepsilon} \longrightarrow 1 - \phi'(0) = 0 \quad (\varepsilon \downarrow 0).$$

Every atom of η_n has mass below ε , and the nonatomic part enters Φ_ϕ with coefficient one. Therefore

$$\begin{aligned} \Phi_\phi(\eta_n) &= \|\eta_{n,\text{cont}}\|_{\mathcal{M}(\Omega)} + \sum_{\omega \in \text{atom } \eta_n} \phi(|\eta_n|(\{\omega\})) \\ &\geq (1 - \delta_\varepsilon) \left(\|\eta_{n,\text{cont}}\|_{\mathcal{M}(\Omega)} + \sum_{\omega \in \text{atom } \eta_n} |\eta_n|(\{\omega\}) \right) \\ &= (1 - \delta_\varepsilon) \|\eta_n\|_{\mathcal{M}(\Omega)}. \end{aligned}$$

The weak-* lower semicontinuity of the total-variation norm and the second inequality in (12) imply

$$(15) \quad \liminf_{n \rightarrow \infty} \Phi_\phi(\eta_n) \geq (1 - \delta_\varepsilon) \|\eta\|_{\mathcal{M}(\Omega)} \geq (1 - \delta_\varepsilon) \Phi_\phi(\eta).$$

Finally, Φ_ϕ is subadditive. Indeed, monotonicity and subadditivity of ϕ give, at every atom,

$$\phi(|\zeta + \eta|(\{\omega\})) \leq \phi(|\zeta|(\{\omega\})) + \phi(|\eta|(\{\omega\})),$$

and the total-variation norm is subadditive on the nonatomic part. Hence

$$\Phi_\phi(\mu) = \Phi_\phi(\zeta + \eta) \leq \Phi_\phi(\zeta) + \Phi_\phi(\eta).$$

Combining (13), (14), and (15) yields

$$\begin{aligned} \ell &\geq \Phi_\phi(\zeta) + (1 - \delta_\varepsilon) \Phi_\phi(\eta) \\ &\geq (1 - \delta_\varepsilon) [\Phi_\phi(\zeta) + \Phi_\phi(\eta)] \end{aligned}$$

$$\geq (1 - \delta_\varepsilon) \Phi_\phi(\mu).$$

Letting $\varepsilon \downarrow 0$ proves $\Phi_\phi(\mu) \leq \ell$. $\square$

For $\alpha > 0$, define

$$(16) \quad J_0(\mu) := L(\mu) + \alpha \Phi_\phi(\mu)$$

for the problem without the moment term.

3.3. Escape on the unbounded parameter domain. For positively homogeneous profiles the radial scale of ω is absorbed by the outer coefficient, and the parameter can be restricted to the unit sphere $\mathbb{S}^d$ (Section 4). But in general the domain is $\Omega = \mathbb{R}^{d+1}$, which is not compact. The functional J_0 need not be well posed. The regularizer Φ_ϕ depends on the sizes of the coefficients but not on the locations of the atoms. Along a sequence with $|\omega| \rightarrow \infty$, the fidelity need not be lower semicontinuous.

Example 3.10 (Failure of weak-* lower semicontinuity for tanh). Let $\rho = \tanh$, $V \equiv 1$, fix $a \in \mathbb{R}^d$, and set $\mu_n = \delta_{(a, b_n)}$ with $b_n \rightarrow \infty$. Then $\mu_n \xrightarrow{*} 0$, yet

$$\mathcal{N}\mu_n = \tanh(a \cdot x + b_n) \rightarrow 1$$

strongly in $H^1(D)$. Hence $\liminf_{n \rightarrow \infty} L(\mu_n) = 0 < L(0)$, so the fidelity term is not weak-* lower semicontinuous. Mass escaping to infinity carries a non-vanishing value contribution unless $K(\omega)$ tends to zero as $|\omega| \rightarrow \infty$. Note that $\Phi_\phi(\mu_n) = \phi(1)$ is constant along this sequence, so the sequence has no narrow limit. Indeed, any narrow limit would also be its weak-* limit and would therefore equal zero. Testing narrow convergence against the constant function 1 would instead give $\mu(\Omega) = \lim_n \mu_n(\Omega) = 1$, which is a contradiction. This is the compactness failure that the moment bound removes.

Example 3.11 (Nonattainment for softplus). For the profile $\rho(z) = \log(1 + e^z)$ (softplus), let $b_n \rightarrow \infty$. The measures $\mu_n = \rho(b_n)^{-1} \delta_{(0, b_n)}$ fit $V \equiv 1$ exactly for every n at penalty $\alpha \phi(1/\rho(b_n)) \rightarrow 0$, so $\inf J_0 = 0$ is not attained.

The boundary behavior can also differ from the interior behavior. Let $F = \partial D \cap \{x : n_F \cdot x = c_F\}$ be a flat face of positive surface measure, assume that D lies on one side of its supporting hyperplane, and let $\rho \in C_0(\mathbb{R})$. Choose z_* with $\rho(z_*) \neq 0$, let $t_j \rightarrow \infty$, and set

$$a_j = t_j n_F, \quad b_j = -t_j c_F + z_*.$$

Then

$$[K(a_j, b_j)](x) = \rho(t_j(n_F \cdot x - c_F) + z_*).$$

For almost every $x \in D$, the argument tends to one of the two infinities. Since ρ is bounded and vanishes at infinity, dominated convergence gives $K(a_j, b_j) \rightarrow 0$ in $L^2(D)$. On the face F , however, $[K(a_j, b_j)]|_F = \rho(z_*)$ for every j . The resulting boundary term is evaluated explicitly below, and the corresponding escaping directions are characterized in Corollary 3.27.

3.4. The moment-regularized problem. For $p > 0$, the space $\mathcal{M}_p(\Omega)$ and its norm are defined in (4). The norm controls both total variation and the locations of the measure:

$$\|\mu\|_{\mathcal{M}(\Omega)} \leq \|\mu\|_{\mathcal{M}_p(\Omega)}.$$

Lemma 3.12 (Compactness from a moment bound). *If $(\mu_n) \subset \mathcal{M}_p(\Omega)$ and $\sup_n \|\mu_n\|_{\mathcal{M}_p(\Omega)} \leq C$, then (μ_n) has a narrowly convergent subsequence.*

Proof. Since $w_p \geq 1$, $\sup_n \|\mu_n\|_{\mathcal{M}(\Omega)} \leq C$. Moreover, for every $R > 0$,

$$|\mu_n|(\Omega \setminus B_R) = \int_{\Omega \setminus B_R} 1 d|\mu_n|$$

$$\begin{aligned} &\leq \frac{1}{1+R^p} \int_{\Omega \setminus B_R} (1 + |\omega|^p) d|\mu_n| \\ &\leq \frac{C}{1+R^p}. \end{aligned}$$

Given $\varepsilon > 0$, choose R so that $C/(1+R^p) < \varepsilon$. Thus $(|\mu_n|)$ is uniformly tight, and Corollary 2.4 applies. $\square$

Lemma 3.13 (Lower semicontinuity of the moment norm). *If $\mu_n \rightharpoonup_n \mu$, then*

$$\|\mu\|_{\mathcal{M}_p(\Omega)} \leq \liminf_{n \rightarrow \infty} \|\mu_n\|_{\mathcal{M}_p(\Omega)}.$$

Proof. Let $g_j := \min\{w_p, j\}$. For every $\tilde{\mu} \in \mathcal{M}(\Omega)$,

$$\int_{\Omega} g_j d|\tilde{\mu}| = \sup_{\substack{f \in C_0(\Omega) \\ \|f\|_{\infty} \leq 1}} \int_{\Omega} f g_j d\tilde{\mu}.$$

Since $f g_j \in C_0(\Omega) \subset C_b(\Omega)$, narrow convergence gives

$$\begin{aligned} \int_{\Omega} g_j d|\mu| &= \sup_{\substack{f \in C_0(\Omega) \\ \|f\|_{\infty} \leq 1}} \lim_{n \rightarrow \infty} \int_{\Omega} f g_j d\mu_n \\ &\leq \liminf_{n \rightarrow \infty} \sup_{\substack{f \in C_0(\Omega) \\ \|f\|_{\infty} \leq 1}} \int_{\Omega} f g_j d\mu_n \\ &= \liminf_{n \rightarrow \infty} \int_{\Omega} g_j d|\mu_n| \\ &\leq \liminf_{n \rightarrow \infty} \|\mu_n\|_{\mathcal{M}_p(\Omega)}. \end{aligned}$$

The second line uses only the following one-sided inequality. For every fixed f ,

$$\lim_{n \rightarrow \infty} \int_{\Omega} f g_j d\mu_n \leq \liminf_{n \rightarrow \infty} \sup_{\substack{h \in C_0(\Omega) \\ \|h\|_{\infty} \leq 1}} \int_{\Omega} h g_j d\mu_n.$$

Taking the supremum over f preserves this inequality; no interchange of a supremum and a limit is required. Letting $j \rightarrow \infty$ and using monotone convergence proves the claim. $\square$

For $\alpha, \beta > 0$, define

$$(17) \quad J(\mu) := L(\mu) + \alpha \Phi_{\phi}(\mu) + \beta \|\mu\|_{\mathcal{M}_p(\Omega)}, \quad \mu \in \mathcal{M}_p(\Omega),$$

and extend J by $+\infty$ outside $\mathcal{M}_p(\Omega)$.

Definition 3.14 (Local minimizer). A measure $\bar{\mu} \in \mathcal{M}_p(\Omega)$ is a local minimizer of J if there is $\varepsilon > 0$ such that

$$J(\bar{\mu}) \leq J(\mu) \quad \text{whenever} \quad \mu \in \mathcal{M}_p(\Omega), \quad \|\mu - \bar{\mu}\|_{\mathcal{M}(\Omega)} \leq \varepsilon.$$

The perturbation is measured in the total-variation norm, which is weaker than the $\mathcal{M}_p(\Omega)$ norm. Its neighborhoods are therefore larger, so Definition 3.14 is the stronger of the two corresponding notions of local minimality.

3.5. Existence of a global minimizer. We apply the direct method of the calculus of variations using narrow convergence. The moment bound gives compactness, the two regularizers are lower semicontinuous, and the following lemma gives the continuity needed to pass to the limit in the fidelity.

Lemma 3.15. *Assume (8), let $p > s_0$, and suppose $\mu_n \rightharpoonup_n \mu$ with $\sup_n \|\mu_n\|_{\mathcal{M}_p(\Omega)} \leq C$. Then $\mathcal{N}_{\mu_n} \rightarrow \mathcal{N}_{\mu}$ strongly in $L^2(D)$.*

Proof. Lemma 3.13 gives $\|\mu\|_{\mathcal{M}_p(\Omega)} \leq C$. Choose $\chi_R \in C_c(\Omega)$ with $0 \leq \chi_R \leq 1$, $\chi_R = 1$ on B_R and $\text{supp } \chi_R \subset B_{R+1}$. The annulus $B_{R+1} \setminus B_R$ allows χ_R to decrease continuously from 1 to 0 while remaining compactly supported. For fixed R and $x \in D$, the function

$$\omega \mapsto \chi_R(\omega)[K(\omega)](x)$$

is bounded and continuous. Hence narrow convergence gives pointwise convergence

$$\int_{\Omega} \chi_R(\omega)[K(\omega)](x) d\mu_n(\omega) \longrightarrow \int_{\Omega} \chi_R(\omega)[K(\omega)](x) d\mu(\omega).$$

Because D is bounded and $\text{supp } \chi_R$ is compact, the pre-activations $a \cdot x + b$ lie in a fixed compact interval. Thus

$$M_R := \sup_{\omega \in \text{supp } \chi_R, x \in D} |[K(\omega)](x)| < \infty.$$

The total variations of μ_n and μ are bounded by C , and hence

$$\left| \int_{\Omega} \chi_R(\omega)[K(\omega)](x) d\mu_n(\omega) \right| \leq CM_R, \quad \left| \int_{\Omega} \chi_R(\omega)[K(\omega)](x) d\mu(\omega) \right| \leq CM_R.$$

The constant function $2CM_R$ belongs to $L^2(D)$ because D is bounded. Dominated convergence on D therefore yields

$$\int_{\Omega} \chi_R K d\mu_n \longrightarrow \int_{\Omega} \chi_R K d\mu \quad \text{strongly in } L^2(D).$$

To remove the cutoff, let $r := |\omega| > R \geq 1$. Since $p > s_0$,

$$\begin{aligned} (1+r)^{s_0} &\leq (2r)^{s_0} = 2^{s_0} r^{s_0-p} r^p \\ &\leq 2^{s_0} R^{s_0-p} r^p \leq 2^{s_0} R^{s_0-p} (1+r^p). \end{aligned}$$

The middle inequality follows from $s_0 - p < 0$ and $r > R$. Therefore, using (8),

$$\left\| \int_{\Omega} (1 - \chi_R) K d\mu_n \right\|_{L^2(D)} \leq \int_{\Omega \setminus B_R} \|K(\omega)\|_{L^2(D)} d|\mu_n|(\omega) \leq 2^{s_0} C_D C R^{s_0-p},$$

and the same bound holds for μ . Letting $n \rightarrow \infty$ in the truncated integrals and then $R \rightarrow \infty$, and using $s_0 - p < 0$, gives the claim. $\square$

Theorem 3.16 (Existence on the unbounded parameter domain). *Let $D \subset \mathbb{R}^d$ be bounded and Lipschitz, let $V \in H^1(D)$, let ϕ satisfy Assumptions 3.2 and 3.3, and let $\rho \in C(\mathbb{R})$ satisfy (7) for some $s_0 \geq 0$. If $p > s_0$, $\alpha > 0$ and $\beta > 0$, then*

$$\inf_{\mu \in \mathcal{M}_p(\Omega)} J(\mu)$$

admits at least one global minimizer.

Proof. Since $V \in H^1(D)$ we have $J(0) = \frac{1}{2} \|V\|_{H^1(D)}^2 := C < \infty$. Choose a minimizing sequence $(\mu_n) \subset \mathcal{M}_p(\Omega)$ with $J(\mu_n) \leq C$. All three terms of (17) are nonnegative, so $\|\mu_n\|_{\mathcal{M}_p(\Omega)} \leq C/\beta$.

Lemma 3.12 gives, after extracting a subsequence, $\mu_n \rightharpoonup_n \bar{\mu}$, and Lemma 3.13 gives $\bar{\mu} \in \mathcal{M}_p(\Omega)$ with $\|\bar{\mu}\|_{\mathcal{M}_p(\Omega)} \leq \liminf_n \|\mu_n\|_{\mathcal{M}_p(\Omega)}$. Lemma 3.15 upgrades this to

$$(18) \quad \mathcal{N}\mu_n \longrightarrow \mathcal{N}\bar{\mu} \quad \text{strongly in } L^2(D).$$

The bound on the fidelity gives $\sup_n \|\nabla(\mathcal{N}\mu_n)\|_{L^2(D)} \leq \|\nabla V\|_{L^2(D)} + \sqrt{2C}$, passing to a subsequence we have $\nabla(\mathcal{N}\mu_n) \rightharpoonup g$ weakly in $L^2(D)^d$. For every $\psi \in C_c^\infty(D)^d$,

$$\int_D g \cdot \psi dx = \lim_{n \rightarrow \infty} \int_D \nabla(\mathcal{N}\mu_n) \cdot \psi dx = - \lim_{n \rightarrow \infty} \int_D \mathcal{N}\mu_n \text{div } \psi dx = - \int_D \mathcal{N}\bar{\mu} \text{div } \psi dx,$$

the last step by (18). Hence $g = \nabla(\mathcal{N}\bar{\mu})$ distributionally and $\mathcal{N}\bar{\mu} \in H^1(D)$.

The value part of the fidelity is continuous by (18) and the gradient part is weakly lower semicontinuous, so $L(\bar{\mu}) \leq \liminf_n L(\mu_n)$. Lemmas 3.9 and 3.13 give the same inequality for the two regularizers. Combining the three terms, $J(\bar{\mu}) \leq \liminf_n J(\mu_n) = \inf J$. $\square$

Remark 3.17 (The gradient operator need not be bounded on $\mathcal{M}(\Omega)$). Assume that $\rho \in C^1(\mathbb{R})$ is nonconstant. Then there are $z_0 \in \mathbb{R}$, $c > 0$, and $\delta > 0$ such that $|\rho'(z)| \geq c$ for $|z - z_0| \leq \delta$. Choose a ball contained in D , a unit vector e , and a point x_0 at its center. For

$$a_t = te, \quad b_t = z_0 - t(e \cdot x_0),$$

one has

$$\nabla K(a_t, b_t)(x) = te \rho'(t(e \cdot (x - x_0)) + z_0).$$

Inside the ball, the slab $|e \cdot (x - x_0)| \leq \delta/t$ has measure bounded below by C/t for large t . Therefore

$$\|\nabla K(a_t, b_t)\|_{L^2(D)}^2 \geq c^2 t^2 \frac{C}{t} = c^2 C t \rightarrow \infty.$$

Since $\|\delta_{(a_t, b_t)}\|_{\mathcal{M}(\Omega)} = 1$, the gradient operator is not bounded from $\mathcal{M}(\Omega)$ to $L^2(D)^d$ in general. This does not preclude boundedness on $\mathcal{M}_p(\Omega)$ when the growth of $\|\nabla K(\omega)\|_{L^2(D)}$ is controlled by $w_p(\omega)$.

3.6. Optimality conditions and finite support. Let $\bar{\mu}$ be a local minimizer. For a measure μ and a direction μ' , the Gâteaux derivative is

$$(19) \quad DL(\mu)[\mu'] := \lim_{t \rightarrow 0} \frac{L(\mu + t\mu') - L(\mu)}{t} = \langle r_\mu, \mathcal{N}\mu' \rangle_{H^1(D)}.$$

At a local minimizer, define

$$\bar{P}(\omega) := DL(\bar{\mu})[\delta_\omega] = \langle r_{\bar{\mu}}, K(\omega) \rangle_{H^1(D)}.$$

Thus $\bar{P}$ is the function representing the Gâteaux derivative. For differentiable ρ ,

$$(20) \quad \bar{P}(a, b) = \int_D r_{\bar{\mu}}(x) \rho(a \cdot x + b) dx + \int_D \nabla r_{\bar{\mu}}(x) \cdot a \rho'(a \cdot x + b) dx.$$

The second term carries the factor a , so on the unbounded domain $\bar{P}$ is defined pointwise but need not be bounded.

Lemma 3.18 (Boundary representation of $\bar{P}$). *Let $\rho \in C^1(\mathbb{R}) \cap L^\infty(\mathbb{R})$ and assume that $r_{\bar{\mu}} \in H^2(D)$. Then:*

(1) *For every $\omega \in \Omega$,*

$$\bar{P}(\omega) = \langle r_{\bar{\mu}} - \Delta r_{\bar{\mu}}, K(\omega) \rangle_{L^2(D)} + \langle \partial_n r_{\bar{\mu}}, K(\omega) \rangle_{L^2(\partial D)}.$$

In particular, $\bar{P}$ is continuous and bounded, with

$$\sup_{\omega \in \Omega} |\bar{P}(\omega)| \leq C_D \|\rho\|_{L^\infty(\mathbb{R})} \|r_{\bar{\mu}}\|_{H^2(D)}.$$

(2) *For every $\mu' \in \mathcal{M}(\Omega)$ such that $\mathcal{N}\mu' \in H^1(D)$,*

$$DL(\bar{\mu})[\mu'] = \int_\Omega \bar{P}(\omega) d\mu'(\omega).$$

Proof. For fixed $\omega = (a, b)$, the function $K(\omega)(x) = \rho(a \cdot x + b)$ belongs to $C^1(\overline{D}) \subset H^1(D)$ and satisfies

$$|K(\omega)(x)| \leq \|\rho\|_{L^\infty(\mathbb{R})} \quad (x \in \overline{D}).$$

Green's formula for $r_{\bar{\mu}} \in H^2(D)$ and $K(\omega) \in H^1(D)$ gives

$$\langle \nabla r_{\bar{\mu}}, \nabla K(\omega) \rangle_{L^2(D)} = -\langle \Delta r_{\bar{\mu}}, K(\omega) \rangle_{L^2(D)} + \langle \partial_n r_{\bar{\mu}}, K(\omega) \rangle_{L^2(\partial D)}.$$

Adding $\langle r_{\bar{\mu}}, K(\omega) \rangle_{L^2(D)}$ proves the representation in part (1). By Cauchy–Schwarz,

$$\begin{aligned} |\bar{P}(\omega)| &\leq \|r_{\bar{\mu}} - \Delta r_{\bar{\mu}}\|_{L^2(D)} \|K(\omega)\|_{L^2(D)} \\ &\quad + \|\partial_n r_{\bar{\mu}}\|_{L^2(\partial D)} \|K(\omega)\|_{L^2(\partial D)}. \end{aligned}$$

The trace theorem applied to $\nabla r_{\bar{\mu}} \in H^1(D)^d$ gives $\partial_n r_{\bar{\mu}} \in L^2(\partial D)$ and

$$\|\partial_n r_{\bar{\mu}}\|_{L^2(\partial D)} \leq C_D \|r_{\bar{\mu}}\|_{H^2(D)}.$$

Together with the uniform bound on $K(\omega)$ in D and on ∂D , this proves the asserted bound after increasing C_D .

If $\omega_j \rightarrow \omega$, then $K(\omega_j) \rightarrow K(\omega)$ pointwise on $\bar{D}$. The uniform bound by $\|\rho\|_\infty$ permits dominated convergence both in D and on ∂D . The representation in part (1) therefore gives $\bar{P}(\omega_j) \rightarrow \bar{P}(\omega)$.

For part (2), let $\mu' \in \mathcal{M}(\Omega)$ satisfy $u := \mathcal{N}\mu' \in H^1(D)$. The trace operator on a bounded Lipschitz domain is bounded,

$$\text{Tr} : H^1(D) \longrightarrow H^{1/2}(\partial D) \hookrightarrow L^2(\partial D).$$

Green's formula applied to $r_{\bar{\mu}}$ and u yields

$$\begin{aligned} DL(\bar{\mu})[\mu'] &= \langle r_{\bar{\mu}}, u \rangle_{L^2(D)} + \langle \nabla r_{\bar{\mu}}, \nabla u \rangle_{L^2(D)} \\ (21) \quad &= \langle r_{\bar{\mu}} - \Delta r_{\bar{\mu}}, u \rangle_{L^2(D)} + \langle \partial_n r_{\bar{\mu}}, \text{Tr } u \rangle_{L^2(\partial D)}. \end{aligned}$$

Since ρ is bounded and μ' is finite, Fubini's theorem gives

$$(22) \quad \langle r_{\bar{\mu}} - \Delta r_{\bar{\mu}}, \mathcal{N}\mu' \rangle_{L^2(D)} = \int_{\Omega} \langle r_{\bar{\mu}} - \Delta r_{\bar{\mu}}, K(\omega) \rangle_{L^2(D)} d\mu'(\omega).$$

The same boundedness and dominated convergence show that $\mathcal{N}\mu'$ has the continuous representative

$$[\mathcal{N}\mu'](x) = \int_{\Omega} [K(\omega)](x) d\mu'(\omega), \quad x \in \bar{D}.$$

For a function in $H^1(D) \cap C(\bar{D})$, its Sobolev trace agrees with its classical boundary restriction. Thus, for $\mathcal{H}^{d-1}$ -almost every $x \in \partial D$,

$$[\text{Tr } u](x) = \int_{\Omega} [K(\omega)](x) d\mu'(\omega).$$

Fubini's theorem on ∂D now gives

$$(23) \quad \langle \partial_n r_{\bar{\mu}}, \text{Tr } u \rangle_{L^2(\partial D)} = \int_{\Omega} \langle \partial_n r_{\bar{\mu}}, K(\omega)|_{\partial D} \rangle_{L^2(\partial D)} d\mu'(\omega).$$

Substituting (22) and (23) into (21), and using part (1), proves

$$DL(\bar{\mu})[\mu'] = \int_{\Omega} \bar{P}(\omega) d\mu'(\omega).$$

□

The representation separates an interior term from the boundary term

$$\langle \partial_n r_{\bar{\mu}}, K(\omega)|_{\partial D} \rangle_{L^2(\partial D)} = \int_{\partial D} \partial_n r_{\bar{\mu}}(x) \rho(a \cdot x + b) d\mathcal{H}^{d-1}(x).$$

It explains how $\bar{P}$ can remain nonzero along directions with $|\omega| \rightarrow \infty$ when the traces of $K(\omega)$ do not tend to zero.

The growth bound is the following standing assumption on the full atom.

Assumption 3.19 (Continuity and polynomial growth in $H^1(D)$). The map $K : \Omega \rightarrow H^1(D)$ is continuous and, for some $s_1 \geq 0$,

$$(24) \quad \|K(\omega)\|_{H^1(D)} \leq C_K (1 + |\omega|)^{s_1} \quad (\omega \in \Omega).$$

By Cauchy–Schwarz, (24) gives

$$(25) \quad |\bar{P}(\omega)| \leq \|r_{\bar{\mu}}\|_{H^1(D)} \|K(\omega)\|_{H^1(D)} \leq C_P(1 + |\omega|)^{s_1}, \quad C_P := C_K \|r_{\bar{\mu}}\|_{H^1(D)}.$$

The constant C_P depends on the fixed minimizer through its residual, but not on ω . If $p > s_1$, then

$$\frac{\bar{P}}{w_p} \in C_0(\Omega), \quad \text{hence} \quad \bar{P} \in C_{w_p^{-1}}(\Omega).$$

Consequently, for every $\mu' \in \mathcal{M}_p(\Omega)$,

$$DL(\bar{\mu})[\mu'] = \int_{\Omega} \bar{P} d\mu'.$$

The pointwise definition of $\bar{P}$ as a Gâteaux derivative does not require $\bar{P} \in C_0(\Omega)$.

Theorem 3.20 (The optimality condition). *Let ϕ satisfy Assumptions 3.2 and 3.3, let $\bar{\mu} \in \mathcal{M}_p(\Omega)$ be a local minimizer of J with $J(\bar{\mu}) < \infty$, and let Assumption 3.19 hold. Then*

$$(26) \quad |\bar{P}(\omega)| \leq \alpha + \beta w_p(\omega) \quad \text{for every } \omega \in \Omega,$$

$$(27) \quad \bar{P}(\omega) = -[\alpha \phi'(|\bar{\mu}|(\{\omega\})) + \beta w_p(\omega)] \operatorname{sign}(\bar{\mu})(\omega), \quad |\bar{\mu}|\text{-almost everywhere.}$$

Proof. We use the Gâteaux derivative $DL(\bar{\mu})[\mu']$, not a $\mathcal{M}(\Omega)$ – $C_0(\Omega)$ duality pairing. For a single atom,

$$DL(\bar{\mu})[\delta_{\omega}] = \bar{P}(\omega)$$

because $\mathcal{N}\delta_{\omega} = K(\omega)$.

Let μ' be a compactly supported direction such that $\mathcal{N}\mu' \in H^1(D)$. Then $\mu' \in \mathcal{M}_p(\Omega)$, and local minimality gives, for all sufficiently small $t > 0$,

$$\begin{aligned} 0 &\leq J(\bar{\mu} + t\mu') - J(\bar{\mu}) \\ &= L(\bar{\mu} + t\mu') - L(\bar{\mu}) + \alpha[\Phi_{\phi}(\bar{\mu} + t\mu') - \Phi_{\phi}(\bar{\mu})] \\ &\quad + \beta[\|\bar{\mu} + t\mu'\|_{\mathcal{M}_p(\Omega)} - \|\bar{\mu}\|_{\mathcal{M}_p(\Omega)}]. \end{aligned}$$

We apply this inequality to three types of directions.

1. Let $\omega \in \operatorname{atom} \bar{\mu}$ and $c := \bar{\mu}(\{\omega\}) \neq 0$. For $s \in \{-1, 1\}$, take $\mu' = s\delta_{\omega}$. Since c is nonzero, for sufficiently small t

$$\begin{aligned} \frac{\phi(|c + ts|) - \phi(|c|)}{t} &\longrightarrow s\phi'(|c|) \operatorname{sign}(c), \\ \frac{|c + ts| - |c|}{t} &\longrightarrow s \operatorname{sign}(c). \end{aligned}$$

After division by t and passage to the limit, local minimality gives

$$0 \leq s \{ \bar{P}(\omega) + [\alpha \phi'(|c|) + \beta w_p(\omega)] \operatorname{sign}(c) \}.$$

Using both values of s yields

$$(28) \quad \bar{P}(\omega) = -[\alpha \phi'(|c|) + \beta w_p(\omega)] \operatorname{sign}(c).$$

Since ϕ is nondecreasing and concave with $\phi'(0) = 1$, $0 \leq \phi'(|c|) \leq 1$. Thus $|\bar{P}(\omega)| \leq \alpha + \beta w_p(\omega)$ at every atom.

2. Let $\omega \notin \operatorname{atom} \bar{\mu}$. Adding either $t\delta_{\omega}$ or $-t\delta_{\omega}$ creates a new atom of mass t . For $s \in \{-1, 1\}$,

$$\begin{aligned} \Phi_{\phi}(\bar{\mu} + ts\delta_{\omega}) - \Phi_{\phi}(\bar{\mu}) &= \phi(t), \\ \|\bar{\mu} + ts\delta_{\omega}\|_{\mathcal{M}_p(\Omega)} - \|\bar{\mu}\|_{\mathcal{M}_p(\Omega)} &= tw_p(\omega). \end{aligned}$$

Dividing the variational inequality by t , using $\phi(t)/t \rightarrow \phi'(0) = 1$, and letting $t \downarrow 0$ gives

$$0 \leq s\bar{P}(\omega) + \alpha + \beta w_p(\omega).$$

The two choices of s imply

$$|\bar{P}(\omega)| \leq \alpha + \beta w_p(\omega).$$

Together with the estimate at the existing atoms, this proves (26).

3. For $R > 0$, set

$$\bar{\mu}_{\text{cont}}^R := \bar{\mu}_{\text{cont}}|_{\bar{B}_R}.$$

Continuity of $K : \Omega \rightarrow H^1(D)$ makes K uniformly bounded on $\bar{B}_R$. Therefore it is Bochner integrable against $\bar{\mu}_{\text{cont}}^R$, and $\mathcal{N}\bar{\mu}_{\text{cont}}^R \in H^1(D)$. In particular, the directions $\pm\bar{\mu}_{\text{cont}}^R$ are admissible. Moreover,

$$\begin{aligned} DL(\bar{\mu})[\bar{\mu}_{\text{cont}}^R] &= \left\langle r_{\bar{\mu}}, \int_{\bar{B}_R} K(\omega) d\bar{\mu}_{\text{cont}}(\omega) \right\rangle_{H^1(D)} \\ &= \int_{\bar{B}_R} \bar{P}(\omega) d\bar{\mu}_{\text{cont}}(\omega). \end{aligned}$$

For $s \in \{-1, 1\}$ and $0 < t < 1$, the perturbation $\bar{\mu} + ts\bar{\mu}_{\text{cont}}^R$ multiplies the nonatomic density on $\bar{B}_R$ by $1 + ts > 0$. Hence

$$\begin{aligned} \Phi_\phi(\bar{\mu} + ts\bar{\mu}_{\text{cont}}^R) - \Phi_\phi(\bar{\mu}) &= ts |\bar{\mu}_{\text{cont}}|(\bar{B}_R), \\ \|\bar{\mu} + ts\bar{\mu}_{\text{cont}}^R\|_{\mathcal{M}_p(\Omega)} - \|\bar{\mu}\|_{\mathcal{M}_p(\Omega)} &= ts \int_{\bar{B}_R} w_p d|\bar{\mu}_{\text{cont}}|. \end{aligned}$$

Dividing by t and using both signs gives

$$(29) \quad \int_{\bar{B}_R} [\bar{P}(\omega) \text{sign}(\bar{\mu}_{\text{cont}})(\omega) + \alpha + \beta w_p(\omega)] d|\bar{\mu}_{\text{cont}}|(\omega) = 0.$$

The nonatomic measure gives zero mass to atom $\bar{\mu}$. Therefore (26) holds $|\bar{\mu}_{\text{cont}}|$ -almost everywhere, and the integrand in (29) is nonnegative:

$$\bar{P} \text{sign}(\bar{\mu}_{\text{cont}}) + \alpha + \beta w_p \geq -|\bar{P}| + \alpha + \beta w_p \geq 0.$$

It must consequently vanish $|\bar{\mu}_{\text{cont}}|$ -almost everywhere on $\bar{B}_R$. Letting $R \rightarrow \infty$ gives

$$(30) \quad \bar{P}(\omega) = -[\alpha + \beta w_p(\omega)] \text{sign}(\bar{\mu}_{\text{cont}})(\omega) \quad |\bar{\mu}_{\text{cont}}| \text{-almost everywhere.}$$

At an atom, $|\bar{\mu}|(\{\omega\}) = |c|$ and $\text{sign}(\bar{\mu})(\omega) = \text{sign}(c)$, so (28) gives (27). On the nonatomic part, $|\bar{\mu}|(\{\omega\}) = 0$ and $\phi'(0) = 1$, so (30) gives the same formula. This completes the proof. $\square$

The conditions (26) and (27) give the threshold $\alpha + \beta w_p(\omega)$. Comparing its growth with (25) gives a bound on the support.

Lemma 3.21 (Boundedness of the support). *Let ϕ satisfy Assumptions 3.2 and 3.3, let $\bar{\mu}$ be a local minimizer of J with $J(\bar{\mu}) < \infty$, let Assumption 3.19 hold, and let $p > s_1$. Then*

$$\text{supp } \bar{\mu} \subset \bar{B}_{R_*}, \quad R_* := \max \left\{ 1, \left(\frac{2^{s_1} C_P}{\beta} \right)^{1/(p-s_1)} \right\},$$

with $C_P = C_K \|r_{\bar{\mu}}\|_{H^1(D)}$ as in (25).

Proof. Taking absolute values in (27) and using $\phi' \geq 0$ gives

$$|\bar{P}(\omega)| \geq \beta w_p(\omega) \quad |\bar{\mu}| \text{-almost everywhere.}$$

Hence $|\bar{\mu}|$ is concentrated on

$$E := \{\omega \in \Omega : \beta w_p(\omega) \leq |\bar{P}(\omega)|\},$$

which is closed because $\bar{P}$ and w_p are continuous. Let $\omega \in E$ and $R := |\omega| \geq 1$. Then $w_p(\omega) = 1 + R^p$ and (25) give

$$\beta R^p \leq \beta(1 + R^p) \leq |\bar{P}(\omega)| \leq C_P(1 + R)^{s_1} \leq 2^{s_1} C_P R^{s_1},$$

so $R^{p-s_1} \leq 2^{s_1} C_P / \beta$. Since $p - s_1 > 0$ this fails for large R , whence $E \subset \bar{B}_{R_*}$. As $|\bar{\mu}|(\Omega \setminus E) = 0$ and E is closed, $\text{supp } \bar{\mu} \subset E$. $\square$

Remark 3.22. For $J_0 = L + \alpha\Phi_\phi$, a sufficient asymptotic condition in the finite-support argument is

$$\limsup_{|\omega| \rightarrow \infty} |\bar{P}(\omega)| < \alpha.$$

For $J = L + \alpha\Phi_\phi + \beta\|\cdot\|_{\mathcal{M}_p(\Omega)}$, the corresponding condition is

$$\limsup_{|\omega| \rightarrow \infty} [|\bar{P}(\omega)| - \beta(1 + |\omega|^p)] < \alpha.$$

If $p > s_1$, then (25) gives

$$|\bar{P}(\omega)| - \beta(1 + |\omega|^p) \leq C_P(1 + |\omega|)^{s_1} - \beta(1 + |\omega|^p) \longrightarrow -\infty.$$

Theorem 3.23 (Finite support). *Let ϕ satisfy Assumptions 3.2, 3.3, and 3.5, let Assumption 3.19 hold and let $p > s_1$. Then every local minimizer $\bar{\mu} \in \mathcal{M}_p(\Omega)$ of J with $J(\bar{\mu}) < \infty$ is finitely supported.*

Proof. By Lemma 3.21, $\text{supp } \bar{\mu}$ lies in the compact ball $\bar{B}_{R_*}$; all subsequent perturbations therefore stay in a fixed compact set, on which K is uniformly continuous into $H^1(D)$.

$$(31) \quad \vartheta_K(\delta) := \sup_{\substack{\omega, \omega' \in \bar{B}_{R_*} \\ |\omega - \omega'| \leq \delta}} \|K(\omega) - K(\omega')\|_{H^1(D)} \longrightarrow 0 \quad (\delta \downarrow 0).$$

The continuous part vanishes. Suppose $\bar{\mu}_{\text{cont}}^+ \neq 0$, the negative case being obtained by reversing the signs of the moved mass. By (27), choose $\hat{\omega} \in \text{supp } \bar{\mu}_{\text{cont}}^+$ with $\bar{\mu}(\{\hat{\omega}\}) = 0$ and $\bar{P}(\hat{\omega}) + \beta w_p(\hat{\omega}) = -\alpha$. For $\delta > 0$ put $\mu_\delta^{\text{loc}} := \bar{\mu}_{\text{cont}}^+|_{B_\delta(\hat{\omega})}$, $m_\delta := \mu_\delta^{\text{loc}}(\Omega) > 0$, and merge this mass onto its centre, $\hat{\mu}_\delta := \bar{\mu} - \mu_\delta^{\text{loc}} + m_\delta \delta_{\hat{\omega}}$, so that $\|\hat{\mu}_\delta - \bar{\mu}\|_{\mathcal{M}(\Omega)} \leq 2m_\delta \rightarrow 0$. Moreover,

$$\mathcal{N}(\hat{\mu}_\delta - \bar{\mu}) = \int_{B_\delta(\hat{\omega})} [K(\hat{\omega}) - K(\omega)] d\mu_\delta^{\text{loc}}(\omega).$$

The triangle inequality for the Bochner integral and (31) give

$$\|\mathcal{N}(\hat{\mu}_\delta - \bar{\mu})\|_{H^1(D)} \leq \vartheta_K(\delta)m_\delta.$$

The quadratic fidelity has the exact expansion

$$L(\hat{\mu}_\delta) - L(\bar{\mu}) = \int_{\Omega} \bar{P} d(\hat{\mu}_\delta - \bar{\mu}) + \frac{1}{2} \|\mathcal{N}(\hat{\mu}_\delta - \bar{\mu})\|_{H^1(D)}^2.$$

The linear fidelity term and the change of the moment cancel:

$$\begin{aligned} & \int_{\Omega} \bar{P} d(\hat{\mu}_\delta - \bar{\mu}) + \beta[\|\hat{\mu}_\delta\|_{\mathcal{M}_p(\Omega)} - \|\bar{\mu}\|_{\mathcal{M}_p(\Omega)}] \\ &= m_\delta[\bar{P}(\hat{\omega}) + \beta w_p(\hat{\omega})] - \int_{B_\delta(\hat{\omega})} [\bar{P} + \beta w_p] d\mu_\delta^{\text{loc}} = 0, \end{aligned}$$

where (27) is used in the last equality. The moved mass was continuous and creates one new atom, hence

$$\Phi_\phi(\hat{\mu}_\delta) - \Phi_\phi(\bar{\mu}) = \phi(m_\delta) - m_\delta \leq -\frac{\gamma_2}{2} m_\delta^2$$

for small δ , by Assumption 3.5(2). Consequently,

$$J(\hat{\mu}_\delta) - J(\bar{\mu}) \leq \frac{1}{2} [\vartheta_K(\delta)^2 - \alpha\gamma_2] m_\delta^2 < 0$$

for δ small, contradicting local minimality. Hence $\bar{\mu} = \bar{\mu}_{\text{atom}}$.

The atom set is finite. If not, infinitely many atoms lie in the compact ball $\bar{B}_{R_*}$. Thus there are distinct atoms $\omega_i \rightarrow \hat{\omega}$ whose coefficients have a common sign. Reversing signs if necessary, write $c_i := \bar{\mu}(\{\omega_i\}) > 0$. Finite total variation gives $c_i \rightarrow 0$, and (27), continuity of $\bar{P}$ and w_p , and $\phi'(c_i) \rightarrow \phi'(0) = 1$ give

$$\bar{P}(\hat{\omega}) = -\alpha - \beta w_p(\hat{\omega}).$$

For $j \in \mathbb{N}$ set

$$C_j := \sum_{i=j}^{\infty} c_i, \quad \varepsilon_j := \sup_{i \geq j} |\omega_i - \hat{\omega}|,$$

and merge this tail,

$$\mu_j := \bar{\mu} - \sum_{i=j}^{\infty} c_i \delta_{\omega_i} + C_j \delta_{\hat{\omega}}.$$

Then $C_j \rightarrow 0$, $\varepsilon_j \rightarrow 0$, and $\|\mu_j - \bar{\mu}\|_{\mathcal{M}(\Omega)} \leq 2C_j \rightarrow 0$. Uniform continuity of K on the compact support gives

$$\mathcal{N}(\mu_j - \bar{\mu}) = \sum_{i=j}^{\infty} c_i [K(\hat{\omega}) - K(\omega_i)],$$

$$\|\mathcal{N}(\mu_j - \bar{\mu})\|_{H^1(D)} \leq \vartheta_K(\varepsilon_j) C_j.$$

Whether or not $\hat{\omega}$ is already an atom, subadditivity of ϕ and the triangle inequality give

$$\|\mu_j\|_{\mathcal{M}_p(\Omega)} - \|\bar{\mu}\|_{\mathcal{M}_p(\Omega)} \leq - \sum_{i=j}^{\infty} c_i w_p(\omega_i) + C_j w_p(\hat{\omega}),$$

$$\Phi_\phi(\mu_j) - \Phi_\phi(\bar{\mu}) \leq - \sum_{i=j}^{\infty} \phi(c_i) + \phi(C_j).$$

The exact quadratic expansion of the fidelity gives

$$L(\mu_j) - L(\bar{\mu}) = \int_{\Omega} \bar{P} d(\mu_j - \bar{\mu}) + \frac{1}{2} \|\mathcal{N}(\mu_j - \bar{\mu})\|_{H^1(D)}^2.$$

By (27) at the atoms ω_i and by the value of $\bar{P}(\hat{\omega})$ obtained above,

$$\begin{aligned} \int_{\Omega} \bar{P} d(\mu_j - \bar{\mu}) &= - \sum_{i=j}^{\infty} c_i \bar{P}(\omega_i) + C_j \bar{P}(\hat{\omega}) \\ &= \alpha \sum_{i=j}^{\infty} c_i \phi'(c_i) + \beta \sum_{i=j}^{\infty} c_i w_p(\omega_i) \\ &\quad - \alpha C_j - \beta C_j w_p(\hat{\omega}). \end{aligned}$$

The two terms containing β cancel with the upper bound for the change of the moment norm. Combining the fidelity and both regularizers therefore yields

$$\begin{aligned} J(\mu_j) - J(\bar{\mu}) &\leq -\alpha \sum_{i=j}^{\infty} [\phi(c_i) - c_i \phi'(c_i)] - \alpha [C_j - \phi(C_j)] \\ &\quad + \frac{1}{2} \vartheta_K(\varepsilon_j)^2 C_j^2. \end{aligned}$$

Concavity and $\phi(0) = 0$ give $\phi(c) - c\phi'(c) \geq 0$, while Assumption 3.5(2) gives $C_j - \phi(C_j) \geq \frac{\gamma_2}{2} C_j^2$ for large j . Hence

$$J(\mu_j) - J(\bar{\mu}) \leq \frac{1}{2} [\vartheta_K(\varepsilon_j)^2 - \alpha \gamma_2] C_j^2 < 0$$

for large j , again contradicting local minimality. Thus atom $\bar{\mu}$ is finite. $\square$

Corollary 3.24 (Existence of a finite network). *Under the hypotheses of Theorem 3.16, Assumption 3.19, and Assumption 3.5, if $p > s_1$, then (17) admits a finitely supported global minimizer.*

Proof. Theorem 3.16 gives a global minimizer. It is also a local minimizer in the sense of Definition 3.14, so Theorem 3.23 applies. $\square$

Corollary 3.25 (The insertion subproblem is coercive). *Let $\bar{\mu}$ satisfy the hypotheses of Theorem 3.20, and assume $p > s_1$. Then the map*

$$\omega \mapsto |\bar{P}(\omega)| - \beta w_p(\omega)$$

tends to $-\infty$ as $|\omega| \rightarrow \infty$ and attains its maximum on Ω .

Proof. By (25) the map is bounded above by $C_P(1 + |\omega|)^{s_1} - \beta(1 + |\omega|^p) \rightarrow -\infty$ since $p > s_1$. It is continuous because $K : \Omega \rightarrow H^1(D)$ is continuous, so its superlevel sets are compact and the supremum is attained. $\square$

The corollary gives a maximizer for the insertion problem on the unbounded parameter space. For J_0 , the supremum of $|\bar{P}|$ need not be attained; Example 3.10 gives a sequence with $|\omega| \rightarrow \infty$.

Remark 3.26. The following example shows that $\bar{P}$ need not tend to zero on a domain with a flat face. Let $D = (0, 1)^d$ with $d \geq 2$, let $F = \{x \in \partial D : x_1 = 0\}$, and take $\rho(z) = e^{-z^2}$. Set

$$r_{\bar{\mu}}(x) = \cosh(1 - x_1).$$

Then $r_{\bar{\mu}} \in H^2(D)$ and $\Delta r_{\bar{\mu}} = r_{\bar{\mu}}$, so the interior term $\langle r_{\bar{\mu}} - \Delta r_{\bar{\mu}}, K(\omega) \rangle_{L^2(D)}$ vanishes. The boundary representation of Lemma 3.18 gives

$$\bar{P}(a, b) = \int_{\partial D} \partial_n r_{\bar{\mu}}(x) \rho(a \cdot x + b) d\mathcal{H}^{d-1}(x).$$

On F the outward normal is $-e_1$, so $\partial_n r_{\bar{\mu}} = \sinh(1)$; on the other faces $\partial_n r_{\bar{\mu}} = 0$. Let

$$a_j = t_j e_1, \quad b_j = 0, \quad t_j \rightarrow \infty,$$

so that $|(a_j, b_j)| \rightarrow \infty$. Since $a_j \cdot x + b_j = 0$ on F ,

$$\bar{P}(a_j, b_j) = \sinh(1) \rho(0) \mathcal{H}^{d-1}(F) = \sinh(1).$$

Thus $\bar{P}$ does not decay along this sequence. Its limiting value is $\sinh(1)$, and a model whose insertion threshold is the constant α satisfies the insertion inequality arbitrarily far from the origin whenever $\alpha < \sinh(1)$. With $\beta > 0$, the same sequence does not satisfy the insertion inequality for sufficiently large j , since $\alpha + \beta w_p(\omega_j) \rightarrow \infty$ while $\bar{P}(\omega_j)$ remains bounded. The example does not assert that the displayed $r_{\bar{\mu}}$ is the residual of a local minimizer; it isolates the boundary term. Corollary 3.27 shows that, under H^2 regularity, any sequence with $|\omega_j| \rightarrow \infty$ along which $\bar{P}$ does not vanish must be asymptotically normal to a flat part of the boundary.

Corollary 3.27 (Escaping directions in the H^2 regime). *Assume the hypotheses of Lemma 3.18, and assume in addition that $\rho \in C_0(\mathbb{R})$. Let $\omega_j = (a_j, b_j)$ satisfy $|\omega_j| \rightarrow \infty$ and*

$$\limsup_{j \rightarrow \infty} |\bar{P}(\omega_j)| > 0.$$

Then, after passing to a subsequence,

$$|a_j| \rightarrow \infty, \quad \frac{a_j}{|a_j|} \rightarrow e, \quad \frac{b_j}{|a_j|} \rightarrow -c$$

for some $|e| = 1$ and $c \in \mathbb{R}$ such that

$$\mathcal{H}^{d-1}(\partial D \cap \{x : e \cdot x = c\}) > 0.$$

In particular, if the positive-measure section above is a flat face $F \subset \{x : n_F \cdot x = c_F\}$, then the sequence is asymptotically normal to that face: up to changing the sign of the normal,

$$\frac{a_j}{|a_j|} \rightarrow n_F, \quad \frac{b_j}{|a_j|} \rightarrow -c_F.$$

Proof. Choose a subsequence, not relabeled, and $\varepsilon > 0$ such that $|\bar{P}(\omega_j)| \geq \varepsilon$. By Lemma 3.18(1),

$$|\bar{P}(\omega_j)| \leq (\|r_{\bar{\mu}}\|_{L^2(D)} + \|\Delta r_{\bar{\mu}}\|_{L^2(D)})\kappa(\omega_j) + \|\partial_n r_{\bar{\mu}}\|_{L^2(\partial D)}\kappa_{\partial}(\omega_j),$$

where

$$\kappa(\omega) := \|K(\omega)\|_{L^2(D)}, \quad \kappa_{\partial}(\omega) := \|K(\omega)|_{\partial D}\|_{L^2(\partial D)}.$$

Since $\rho \in C_0(\mathbb{R})$ and D is bounded, $\kappa(\omega_j) \rightarrow 0$: writing $\varepsilon_{\rho}(\tau) := \sup_{|z| \geq \tau} |\rho(z)|$, which tends to 0, the set of $x \in D$ with $|a_j \cdot x + b_j| \leq \tau$ is a slab of width $2\tau/|a_j|$ when $|a_j| \rightarrow \infty$ and is empty for large j when $|a_j|$ stays bounded (so that $|b_j| \rightarrow \infty$); in either case $\limsup_j \kappa(\omega_j)^2 \leq \varepsilon_{\rho}(\tau)^2 |D|$ for every τ .

If $|a_j|$ stayed bounded along a further subsequence, then $|b_j| \rightarrow \infty$, and since ∂D is bounded, $|a_j \cdot x + b_j| \rightarrow \infty$ uniformly on ∂D . Hence $\kappa_{\partial}(\omega_j) \rightarrow 0$, contradicting $|\bar{P}(\omega_j)| \geq \varepsilon$. Therefore $|a_j| \rightarrow \infty$ along the non-vanishing subsequence. Passing to a further subsequence, $a_j/|a_j| \rightarrow e$ with $|e| = 1$.

If $|b_j|/|a_j|$ were unbounded, then again $|a_j \cdot x + b_j| \rightarrow \infty$ uniformly on ∂D , so $\kappa_{\partial}(\omega_j) \rightarrow 0$, a contradiction. Thus, after passing to a further subsequence, $b_j/|a_j| \rightarrow -c$ for some $c \in \mathbb{R}$. Let $\Pi := \{x : e \cdot x = c\}$.

If $\mathcal{H}^{d-1}(\partial D \cap \Pi) = 0$, we show directly that $\kappa_{\partial}(\omega_j) \rightarrow 0$. For $\tau > 0$ set

$$B_j(\tau) := \{x \in \partial D : |a_j \cdot x + b_j| \leq \tau\}, \quad \varepsilon_{\rho}(\tau) := \sup_{|z| \geq \tau} |\rho(z)|.$$

Then

$$\kappa_{\partial}(\omega_j)^2 \leq \|\rho\|_{\infty}^2 \mathcal{H}^{d-1}(B_j(\tau)) + \varepsilon_{\rho}(\tau)^2 \mathcal{H}^{d-1}(\partial D).$$

For $x \in B_j(\tau)$,

$$|e \cdot x - c| \leq \frac{|a_j \cdot x + b_j|}{|a_j|} + \left|e - \frac{a_j}{|a_j|}\right| R_D + \left|c + \frac{b_j}{|a_j|}\right| \rightarrow 0$$

uniformly in x , where $R_D > 0$ satisfies $|x| \leq R_D$ on ∂D . Hence, for every $\varepsilon > 0$, eventually

$$B_j(\tau) \subset \{x \in \partial D : |e \cdot x - c| \leq \varepsilon\}.$$

By continuity from above of $\mathcal{H}^{d-1} \llcorner \partial D$,

$$\limsup_j \mathcal{H}^{d-1}(B_j(\tau)) \leq \mathcal{H}^{d-1}(\partial D \cap \Pi) = 0.$$

Therefore

$$\limsup_j \kappa_{\partial}(\omega_j)^2 \leq \varepsilon_{\rho}(\tau)^2 \mathcal{H}^{d-1}(\partial D)$$

for every $\tau > 0$, and letting $\tau \rightarrow \infty$ gives $\kappa_{\partial}(\omega_j) \rightarrow 0$. Together with $\kappa(\omega_j) \rightarrow 0$, this would imply $\bar{P}(\omega_j) \rightarrow 0$, contradicting the choice of the subsequence. Hence $\mathcal{H}^{d-1}(\partial D \cap \Pi) > 0$.

The last assertion follows because a hyperplane containing a flat face has normal $\pm n_F$ and offset $\pm c_F$, with the signs linked as above. $\square$

For fixed inner parameters $\bar{\omega} = (\bar{\omega}_n)_{n=1}^N$, consider the finite-dimensional problem

$$(32) \quad J_{\bar{\omega}}(c) := L \left(\sum_{n=1}^N c_n \delta_{\bar{\omega}_n} \right) + \alpha \sum_{n=1}^N \phi(|c_n|) + \beta \sum_{n=1}^N w_p(\bar{\omega}_n) |c_n|, \quad c \in \mathbb{R}^N,$$

which is amenable to gradient-based optimization.

We next give a sufficient condition under which a finitely supported candidate is a local minimizer.

Theorem 3.28 (Sufficient condition for local optimality). *Let ϕ satisfy Assumptions 3.2, 3.3, and 3.5, let Assumption 3.19 hold, let $p > s_1$, and let $V \in H^1(D)$. Write*

$$\tau(\omega) := \alpha + \beta w_p(\omega).$$

Let $\bar{\omega} := (\bar{\omega}_n)_{n=1}^N$ and $\bar{\mu} := \sum_{n=1}^N \bar{c}_n \delta_{\bar{\omega}_n}$ with distinct $\bar{\omega}_n$ and $\bar{c}_n \neq 0$, and define

$$\bar{P}(\omega) := \langle \mathcal{N}\bar{\mu} - V, K(\omega) \rangle_{H^1(D)}.$$

Assume that

- (1) $\bar{c} \in (\mathbb{R} \setminus \{0\})^N$ is a local minimum of $J_{\bar{\omega}}$,
- (2) $|\bar{P}(\omega)| < \tau(\omega)$ for all $\omega \in \Omega \setminus \{\bar{\omega}_n\}_{n=1, \dots, N}$.

By (25) and $p > s_1$, hypothesis (2) holds automatically outside a sufficiently large ball and therefore only needs to be checked on a compact set. Then $\bar{\mu}$ is a local minimizer of J : there exists $\epsilon > 0$ such that $J(\bar{\mu}) \leq J(\mu)$ for all $\mu \in \mathcal{M}_p(\Omega)$ with $\|\mu - \bar{\mu}\|_{\mathcal{M}(\Omega)} \leq \epsilon$.

Proof. Step 1: a uniform strict inequality. Since all $\bar{c}_n$ are nonzero, stationarity of $J_{\bar{\omega}}$ at $\bar{c}$ gives

$$\bar{P}(\bar{\omega}_n) = -[\alpha \phi'(|\bar{c}_n|) + \beta w_p(\bar{\omega}_n)] \text{sign}(\bar{c}_n).$$

Assumptions 3.3 and 3.5(2) imply $\phi'(z) < \phi'(0) = 1$ for every $z > 0$. Together with hypothesis (2), this gives $|\bar{P}(\omega)| < \tau(\omega)$ for every $\omega \in \Omega$.

Set $g(\omega) := \tau(\omega) - |\bar{P}(\omega)|$. The function g is continuous because $K : \Omega \rightarrow H^1(D)$ is continuous. Moreover,

$$\frac{g(\omega)}{(1 + |\omega|)^{s_1}} \geq \frac{\alpha + \beta(1 + |\omega|^p)}{(1 + |\omega|)^{s_1}} - C_P \longrightarrow +\infty \quad (|\omega| \rightarrow \infty),$$

by (25) and $p > s_1$. The positive continuous function on the left therefore has a positive infimum: for some $m > 0$,

$$(33) \quad |\bar{P}(\omega)| \leq \tau(\omega) - m(1 + |\omega|)^{s_1} \quad (\omega \in \Omega).$$

Step 2: decomposition of a competitor. Hypothesis (1) gives $\epsilon_1 > 0$ such that $J_{\bar{\omega}}(c) \geq J_{\bar{\omega}}(\bar{c})$ whenever $\|c - \bar{c}\|_{\ell^1} \leq \epsilon_1$. Let

$$C_0 := \max_{1 \leq n \leq N} \|K(\bar{\omega}_n)\|_{H^1(D)}, \quad C_* := C_K C_0.$$

Choose $\epsilon > 0$ with

$$\epsilon \leq \epsilon_1, \quad \epsilon \leq \hat{z}_1, \quad C_* \epsilon \leq \frac{m}{2}, \quad \frac{\alpha \gamma_1 \epsilon}{2} \leq \frac{m}{4}.$$

The last condition is void if $\gamma_1 = 0$. Let $\|\mu - \bar{\mu}\|_{\mathcal{M}(\Omega)} \leq \epsilon$. There is nothing to prove if $J(\mu) = +\infty$. Otherwise $\mu \in \mathcal{M}_p(\Omega)$, and we write

$$\mu = \mu_0 + \tilde{\mu}, \quad \mu_0 := \sum_{n=1}^N c_n \delta_{\bar{\omega}_n}, \quad c_n := \mu(\{\bar{\omega}_n\}).$$

Then $\tilde{\mu}$ carries no mass at the support points and

$$\|c - \bar{c}\|_{\ell^1} + \|\tilde{\mu}\|_{\mathcal{M}(\Omega)} = \|\mu - \bar{\mu}\|_{\mathcal{M}(\Omega)} \leq \epsilon.$$

Step 3: control of the cross term. Write

$$\bar{r} := r_{\bar{\mu}} = \mathcal{N}\bar{\mu} - V, \quad r' := \mathcal{N}(\mu_0 - \bar{\mu}),$$

and define $Q(\omega) := \langle r', K(\omega) \rangle_{H^1(D)}$. Then

$$\begin{aligned} \|r'\|_{H^1(D)} &= \left\| \sum_{n=1}^N (c_n - \bar{c}_n) K(\bar{\omega}_n) \right\|_{H^1(D)} \\ &\leq \sum_{n=1}^N |c_n - \bar{c}_n| \|K(\bar{\omega}_n)\|_{H^1(D)} \leq C_0 \epsilon. \end{aligned}$$

Consequently, Assumption 3.19 and Cauchy–Schwarz give

$$(34) \quad |Q(\omega)| \leq \|r'\|_{H^1(D)} \|K(\omega)\|_{H^1(D)} \leq C_* \epsilon (1 + |\omega|)^{s_1}.$$

Combining this with (33) gives

$$(35) \quad |\bar{P}(\omega) + Q(\omega)| \leq \tau(\omega) - \frac{m}{2}(1 + |\omega|)^{s_1}.$$

Indeed, $C_*\epsilon \leq m/2$ by the choice of ϵ .

Since $J(\mu) < \infty$, the represented function $\mathcal{N}\mu$ belongs to $H^1(D)$. The finite measure μ_0 represents a finite sum of $H^1(D)$ atoms, and hence

$$\mathcal{N}\tilde{\mu} = \mathcal{N}\mu - \mathcal{N}\mu_0 \in H^1(D).$$

Moreover,

$$r_{\mu_0} = \mathcal{N}\mu_0 - V = \bar{r} + r', \quad r_\mu = r_{\mu_0} + \mathcal{N}\tilde{\mu}.$$

Expanding the square in $H^1(D)$ therefore gives the exact identity

$$\begin{aligned} L(\mu) &= \frac{1}{2} \|r_{\mu_0} + \mathcal{N}\tilde{\mu}\|_{H^1(D)}^2 \\ &= L(\mu_0) + \langle r_{\mu_0}, \mathcal{N}\tilde{\mu} \rangle_{H^1(D)} + \frac{1}{2} \|\mathcal{N}\tilde{\mu}\|_{H^1(D)}^2 \\ &= L(\mu_0) + \langle \bar{r}, \mathcal{N}\tilde{\mu} \rangle_{H^1(D)} + \langle r', \mathcal{N}\tilde{\mu} \rangle_{H^1(D)} + \frac{1}{2} \|\mathcal{N}\tilde{\mu}\|_{H^1(D)}^2. \end{aligned}$$

The functions $\bar{P}$ and Q grow at most like $(1 + |\omega|)^{s_1}$. Since $p > s_1$ and $\tilde{\mu} \in \mathcal{M}_p(\Omega)$, they are integrable with respect to $|\tilde{\mu}|$. The Bochner representation of $\mathcal{N}\tilde{\mu}$ then implies

$$\begin{aligned} \langle \bar{r}, \mathcal{N}\tilde{\mu} \rangle_{H^1(D)} &= \int_{\Omega} \bar{P}(\omega) d\tilde{\mu}(\omega), \\ \langle r', \mathcal{N}\tilde{\mu} \rangle_{H^1(D)} &= \int_{\Omega} Q(\omega) d\tilde{\mu}(\omega). \end{aligned}$$

After discarding the nonnegative last term in the square expansion, we obtain

$$(36) \quad L(\mu) \geq L(\mu_0) + \int_{\Omega} [\bar{P}(\omega) + Q(\omega)] d\tilde{\mu}(\omega).$$

Step 4: assembly. The measure μ_0 is supported on $\{\bar{\omega}_1, \dots, \bar{\omega}_N\}$, whereas $\tilde{\mu}$ has zero mass at each of these points. Thus μ_0 and $\tilde{\mu}$ are mutually singular. Their atomic and nonatomic parts are therefore disjoint, and

$$\begin{aligned} \Phi_\phi(\mu) &= \sum_{n=1}^N \phi(|c_n|) + \Phi_\phi(\tilde{\mu}), \\ \|\mu\|_{\mathcal{M}_p(\Omega)} &= \sum_{n=1}^N w_p(\bar{\omega}_n) |c_n| + \|\tilde{\mu}\|_{\mathcal{M}_p(\Omega)}. \end{aligned}$$

Every atom of $\tilde{\mu}$ has mass at most $\|\tilde{\mu}\|_{\mathcal{M}(\Omega)} \leq \epsilon \leq \hat{z}_1$. Proposition 3.7(1) consequently gives

$$\Phi_\phi(\tilde{\mu}) \geq \left(1 - \frac{\gamma_1 \epsilon}{2}\right) \|\tilde{\mu}\|_{\mathcal{M}(\Omega)}.$$

Using (36) and the two splittings above,

$$\begin{aligned} J(\mu) &\geq J_{\bar{\omega}}(c) + \int_{\Omega} [\bar{P} + Q] d\tilde{\mu} + \alpha \Phi_\phi(\tilde{\mu}) + \beta \|\tilde{\mu}\|_{\mathcal{M}_p(\Omega)} \\ &\geq J_{\bar{\omega}}(c) - \int_{\Omega} \left[\alpha + \beta w_p(\omega) - \frac{m}{2}(1 + |\omega|)^{s_1} \right] d|\tilde{\mu}|(\omega) \\ &\quad + \alpha \left(1 - \frac{\gamma_1 \epsilon}{2}\right) \|\tilde{\mu}\|_{\mathcal{M}(\Omega)} + \beta \|\tilde{\mu}\|_{\mathcal{M}_p(\Omega)} \\ &= J_{\bar{\omega}}(c) + \frac{m}{2} \int_{\Omega} (1 + |\omega|)^{s_1} d|\tilde{\mu}|(\omega) - \frac{\alpha \gamma_1 \epsilon}{2} \|\tilde{\mu}\|_{\mathcal{M}(\Omega)} \\ &\geq J_{\bar{\omega}}(c) + \frac{m}{4} \|\tilde{\mu}\|_{\mathcal{M}(\Omega)} \geq J_{\bar{\omega}}(\bar{c}) = J(\bar{\mu}). \end{aligned}$$

Thus $\bar{\mu}$ is a local minimizer. $\square$

3.7. Admissible activations. The two exponents play different roles: s_0 controls the value atom in $L^2(D)$ and is enough for Theorem 3.16; s_1 controls the full atom in $H^1(D)$ and is what Lemma 3.21 and Theorem 3.23 require. Since $\|K(\omega)\|_{L^2(D)} \leq \|K(\omega)\|_{H^1(D)}$, the exponents can be chosen so that $s_0 \leq s_1$. The exponent s_1 is obtained by estimating $a\rho'(a \cdot x + b)$ separately. The condition $p > s_1$ is sufficient for existence and finite support. The following exponents satisfy the pointwise bounds on a bounded domain.

activation	s_0	s_1	sufficient condition
$\tanh z$	0	1	$p > 1$
e^{-z^2}	0	1	$p > 1$
$\log(1 + e^z)$	1	1	$p > 1$
$(z_+)^k$	k	k	$p > k$

For bounded activations such as $\tanh$ and the Gaussian, existence alone requires only $p > 0$; the condition $p > 1$ in the table is the one that also dominates the elementary global H^1 growth estimate. The last row is listed for comparison with the positively homogeneous formulation of Section 4.

4. THE k -HOMOGENEOUS FORMULATION

This section treats activations for which radial scaling can be absorbed into the outer coefficient.

Assumption 4.1 (k -homogeneity). For some $k \geq 1$,

$$K(\lambda\omega) = \lambda^k K(\omega) \quad (\lambda > 0, \omega \in \mathbb{R}^{d+1}).$$

Remark 4.2. For gradient fitting with ReLU^k , we use $k \geq 2$, so $K(\omega) \in C^1(\overline{D})$. Its derivative is, up to the factor k , a ReLU^{k-1} ridge function: the value term contains powers of order k , whereas the gradient term contains powers of order $k-1$. When $k=2$, the gradient is continuous but is not differentiable on the hyperplane $a \cdot x + b = 0$. The algebraic scaling results below hold for every $k \geq 1$.

Proposition 4.3 (Rescaling of a k -homogeneous network). *Let Assumption 4.1 hold, and let $\xi, \zeta \geq 1$. For a fixed represented coefficient $\tilde{c}$ and a unit direction $\hat{\omega} \in \mathbb{S}^d$ with $K(\hat{\omega}) \neq 0$, minimize*

$$\frac{|c|^\xi}{\xi} + \frac{|\omega|^\zeta}{\zeta}$$

over radial representations

$$\omega = \lambda \hat{\omega}, \quad \lambda \geq 0, \quad c K(\omega) = \tilde{c} K(\hat{\omega}).$$

The minimum equals

$$(37) \quad C_{k,\xi,\zeta} |\tilde{c}|^q, \quad q := \frac{\xi\zeta}{k\xi + \zeta}, \quad C_{k,\xi,\zeta} := \frac{1}{q} k^{-k\xi/(k\xi + \zeta)}.$$

Consequently, an objective with the separable penalty above is equivalent to an objective on $\mathbb{S}^d$ with the induced penalty $C_{k,\xi,\zeta} \sum_n |c_n|^q$.

Remark 4.4. For $\xi = \zeta = 2$,

$$q = \frac{2}{k+1}, \quad C_k := C_{k,2,2} = \frac{k+1}{2} k^{-k/(k+1)}.$$

The numerical implementation uses $\alpha \sum_n |c_n|^{2/(k+1)}$ and absorbs C_k into α .

Proof. If $\tilde{c} = 0$, the zero representation $c = 0$, $\omega = 0$ has penalty zero, in agreement with (37). We therefore assume $\tilde{c} \neq 0$.

Write $\omega = \lambda \hat{\omega}$, where $\lambda = |\omega| > 0$ and $\hat{\omega} \in \mathbb{S}^d$. By k -homogeneity,

$$c K(\omega) = c \lambda^k K(\hat{\omega}) = \tilde{c} K(\hat{\omega}).$$

Since $K(\hat{\omega}) \neq 0$, the representation constraint is equivalent to $c \lambda^k = \tilde{c}$. For fixed $\tilde{c}$, the penalty becomes

$$g(\lambda) = \frac{|\tilde{c}|^\xi}{\xi \lambda^{k\xi}} + \frac{\lambda^\zeta}{\zeta}.$$

The function g tends to $+\infty$ as $\lambda \downarrow 0$ and as $\lambda \rightarrow \infty$. Moreover,

$$g'(\lambda) = -k|\tilde{c}|^\xi \lambda^{-k\xi-1} + \lambda^{\zeta-1},$$

so

$$g'(\lambda) = 0 \iff \lambda^{k\xi+\zeta} = k|\tilde{c}|^\xi.$$

This equation has the unique solution

$$\lambda_* = (k|\tilde{c}|^\xi)^{1/(k\xi+\zeta)}.$$

The derivative is negative before λ_* and positive after λ_* , so this point is the global minimizer. The first term at λ_* is

$$\begin{aligned} \frac{|\tilde{c}|^\xi}{\xi \lambda_*^{k\xi}} &= \frac{1}{\xi} |\tilde{c}|^\xi (k|\tilde{c}|^\xi)^{-k\xi/(k\xi+\zeta)} \\ &= \frac{1}{\xi} k^{-k\xi/(k\xi+\zeta)} |\tilde{c}|^{\xi-k\xi^2/(k\xi+\zeta)} \\ &= \frac{1}{\xi} k^{-k\xi/(k\xi+\zeta)} |\tilde{c}|^{\xi\zeta/(k\xi+\zeta)}. \end{aligned}$$

The second term is

$$\begin{aligned} \frac{\lambda_*^\zeta}{\zeta} &= \frac{1}{\zeta} (k|\tilde{c}|^\xi)^{\zeta/(k\xi+\zeta)} \\ &= \frac{k}{\zeta} k^{-k\xi/(k\xi+\zeta)} |\tilde{c}|^{\xi\zeta/(k\xi+\zeta)}. \end{aligned}$$

Since

$$q = \frac{\xi\zeta}{k\xi+\zeta}, \quad \frac{1}{\xi} + \frac{k}{\zeta} = \frac{k\xi+\zeta}{\xi\zeta} = \frac{1}{q},$$

adding the two terms gives

$$\begin{aligned} g(\lambda_*) &= k^{-k\xi/(k\xi+\zeta)} \left(\frac{1}{\xi} + \frac{k}{\zeta} \right) |\tilde{c}|^q \\ &= \frac{1}{q} k^{-k\xi/(k\xi+\zeta)} |\tilde{c}|^q = C_{k,\xi,\zeta} |\tilde{c}|^q. \end{aligned}$$

This calculation is performed independently for every neuron. The rescaling preserves each represented function $c_n K(\omega_n)$, and hence preserves the network and either the population fidelity L or the empirical fidelity l^M . Summing the minimum costs proves the stated equivalence. $\square$

Using only the coefficient penalty $\alpha \sum_n |c_n|^{2/(k+1)}$ on the unrestricted parameter domain would not control radial rescaling: replacing (c_n, ω_n) by $(c_n \lambda^{-k}, \lambda \omega_n)$ preserves the network while its penalty tends to zero as $\lambda \rightarrow \infty$. Proposition 4.3 instead derives the sphere-normalized coefficient penalty from a separable penalty on the inner and outer parameters. After the normalization $|\omega_n| = 1$, the inner-parameter domain is compact and no moment norm is needed.

For ReLU^k with $k \geq 2$, define

$$\psi_k(z) := z^{2/(k+1)}, \quad z \geq 0.$$

Since $\psi'_k(0^+) = +\infty$, this penalty is used directly for finite networks. For $\omega = (\omega_n)_{n=1}^N$ and $\mu_{\omega,c} = \sum_{n=1}^N c_n \delta_{\omega_n}$, set

$$(38) \quad J_{\psi_k}(\omega, c) := L(\mu_{\omega,c}) + \alpha \sum_{n=1}^N \psi_k(|c_n|),$$

$$(39) \quad J_{\psi_k}^M(\omega, c) := l^M(\mu_{\omega,c}) + \alpha \sum_{n=1}^N \psi_k(|c_n|).$$

Proposition 4.5 (Homogeneous reformulation). *Under Assumption 4.1, minimizing either objective over $|\omega_n| \leq 1$ is equivalent to minimizing it over $\omega_n \in \mathbb{S}^d$:*

$$(P_{\psi_k}) \quad \min_{\substack{N \in \mathbb{N}_0, \omega_n \in \mathbb{S}^d \\ c_n \in \mathbb{R}}} J_{\psi_k}(\omega, c).$$

The same statement holds for $J_{\psi_k}^M$.

Proof. First consider a network with $|\omega_n| \leq 1$. By k -homogeneity, $K(0) = 0$, so an atom with $\omega_n = 0$ represents the zero function and may be discarded. For every remaining atom with $0 < |\omega_n| \leq 1$, set

$$\hat{\omega}_n := \frac{\omega_n}{|\omega_n|}, \quad \hat{c}_n := c_n |\omega_n|^k.$$

Then $\hat{\omega}_n \in \mathbb{S}^d$ and

$$\hat{c}_n K(\hat{\omega}_n) = c_n |\omega_n|^k K(\hat{\omega}_n) = c_n K(\omega_n).$$

Summing over n shows that the normalized parameters represent exactly the same function. Their distributional gradients are consequently equal as well, so both L and l^M are unchanged whenever the corresponding pointwise values in l^M are defined.

Since $|\omega_n| \leq 1$, one has $|\hat{c}_n| = |c_n| |\omega_n|^k \leq |c_n|$. The function ψ_k is nondecreasing, and therefore

$$\sum_n \psi_k(|\hat{c}_n|) \leq \sum_n \psi_k(|c_n|).$$

Thus every network feasible in the unit ball has a sphere-feasible representation with no larger objective value. Conversely, every sphere-feasible network is already feasible in the unit ball. The two infima are equal, and the normalization maps a minimizer in the unit ball to a minimizer on the sphere. A sphere minimizer is also a minimizer in the unit ball because it is feasible there and has the common optimal value. $\square$

The ReLU^k function. For $\rho(z) = (z_+)^k$,

$$[K(\omega)](x) = (a \cdot x + b)_+^k, \quad k \geq 2.$$

The power k is a real number: the homogeneity analysis of this section only uses $k \geq 1$, while the Barron–Sobolev embedding of Proposition 2.5 applies to integer k . Since these activations are not polynomials, the corresponding networks are dense in $C(Q)$ for every compact set $Q \subset \mathbb{R}^d$ [16].

$$\nabla K(\omega)(x) = k(a \cdot x + b)_+^{k-1} a.$$

The second-order weak derivative of ReLU does not exist as a function while it is a distribution, and we cannot use it to approximate high-order derivatives. For $k \geq 2$, the first derivative of $(z_+)^k$ is continuous, so the gradient-augmented fidelity is well defined pointwise.

Corollary 4.6 (Optimality conditions for the fractional-exponent penalty). *Let $q = 2/(k+1) \in (0, 1)$, let $V \in H^1(D)$, and assume that $K : \mathbb{S}^d \rightarrow H^1(D)$ is continuous. Let $(\bar{\omega}, \bar{c}) \in (\mathbb{S}^d)^N \times (\mathbb{R} \setminus \{0\})^N$ be locally optimal with respect to the outer coefficients for J_{ψ_k} . Set*

$$\bar{\mu} := \mu_{\bar{\omega}, \bar{c}}, \quad r_{\bar{\mu}} := \mathcal{N}_{\bar{\omega}, \bar{c}} - V, \quad \bar{P}(\omega) := \langle r_{\bar{\mu}}, K(\omega) \rangle_{H^1(D)}.$$

Then

$$(40) \quad \bar{P}(\bar{\omega}_n) + \alpha q |\bar{c}_n|^{q-1} \text{sign}(\bar{c}_n) = 0.$$

If $B := \|\bar{P}\|_{C(\mathbb{S}^d)} > 0$, every nonzero coefficient satisfies

$$(41) \quad |\bar{c}_n| \geq \left(\frac{\alpha q}{B}\right)^{1/(1-q)} =: c_{\min} > 0.$$

Proof. Fix n and let e_n denote the n th coordinate vector in $\mathbb{R}^N$. For $s \in \{-1, 1\}$ and sufficiently small $t > 0$, local optimality in the outer coefficients gives

$$J_{\psi_k}(\bar{\omega}, \bar{c} + tse_n) - J_{\psi_k}(\bar{\omega}, \bar{c}) \geq 0.$$

The perturbed residual is

$$r_{\bar{\mu}} + tsK(\bar{\omega}_n),$$

and hence the exact quadratic expansion of the fidelity is

$$\begin{aligned} L(\mu_{\bar{\omega}, \bar{c} + tse_n}) - L(\bar{\mu}) &= ts \langle r_{\bar{\mu}}, K(\bar{\omega}_n) \rangle_{H^1(D)} + \frac{t^2}{2} \|K(\bar{\omega}_n)\|_{H^1(D)}^2 \\ &= ts \bar{P}(\bar{\omega}_n) + \frac{t^2}{2} \|K(\bar{\omega}_n)\|_{H^1(D)}^2. \end{aligned}$$

Because $\bar{c}_n \neq 0$, the map $z \mapsto |z|^q$ is differentiable at $\bar{c}_n$. After division by t and passage to the limit $t \downarrow 0$, we obtain

$$0 \leq s [\bar{P}(\bar{\omega}_n) + \alpha q |\bar{c}_n|^{q-1} \text{sign}(\bar{c}_n)].$$

The choices $s = 1$ and $s = -1$ give (40).

The continuity of $K : \mathbb{S}^d \rightarrow H^1(D)$ implies the continuity of $\bar{P}(\omega) = \langle r_{\bar{\mu}}, K(\omega) \rangle_{H^1(D)}$. Since $\mathbb{S}^d$ is compact, $B = \|\bar{P}\|_{C(\mathbb{S}^d)}$ is finite. Taking absolute values in (40) gives

$$\alpha q |\bar{c}_n|^{q-1} = |\bar{P}(\bar{\omega}_n)| \leq B.$$

Since $q - 1 = -(1 - q)$ and $B > 0$, this inequality is equivalent to

$$|\bar{c}_n|^{1-q} \geq \frac{\alpha q}{B},$$

which is precisely (41). $\square$

Remark 4.7. The singular derivative $\psi'_k(0^+) = +\infty$ has three consequences.

- (1) Adding a new neuron with coefficient ts , where $s \in \{-1, 1\}$, changes the penalty by αt^q . Since $t^q/t = t^{q-1} \rightarrow \infty$, division by t gives no pointwise condition on $\bar{P}(\omega)$ away from the existing support.
- (2) At a nonzero coefficient, the same singularity gives the positive lower bound (41). Combining this bound with an upper bound on the objective yields the width estimate below.
- (3) The zero network is a local minimizer with respect to the outer coefficients: for sufficiently small coefficients, the term of order $|c|^q$ dominates the linear change of the fidelity. An insertion criterion based on an infinitesimal coefficient perturbation is therefore unavailable for this penalty.

Corollary 4.8 (Existence and width bound). *Let $\alpha > 0$ and $q = 2/(k+1) \in (0, 1)$, assume that $K : \mathbb{S}^d \rightarrow H^1(D)$ is continuous and is not identically zero, and let $V \in H^1(D) \setminus \{0\}$. Then (P_{ψ_k}) admits a global minimizer. After zero coefficients are discarded, its width satisfies*

$$N \leq N_{\max} := \frac{\|V\|_{H^1(D)}^2}{2\alpha c_{\min}^q}, \quad c_{\min} := \left(\frac{\alpha q}{\|V\|_{H^1(D)} \sup_{\omega \in \mathbb{S}^d} \|K(\omega)\|_{H^1(D)}} \right)^{1/(1-q)}.$$

Proof. Step 1: existence at a fixed width. Fix $N \in \mathbb{N}$. We first verify that J_{ψ_k} is continuous on $(\mathbb{S}^d)^N \times \mathbb{R}^N$. Suppose

$$(\omega^{(j)}, c^{(j)}) \longrightarrow (\omega, c).$$

Continuity of $K : \mathbb{S}^d \rightarrow H^1(D)$ and compactness of the sphere give

$$C_S := \sup_{\omega \in \mathbb{S}^d} \|K(\omega)\|_{H^1(D)} < \infty.$$

For every n ,

$$\begin{aligned} & \|c_n^{(j)} K(\omega_n^{(j)}) - c_n K(\omega_n)\|_{H^1(D)} \\ & \leq |c_n^{(j)} - c_n| C_S + |c_n| \|K(\omega_n^{(j)}) - K(\omega_n)\|_{H^1(D)} \longrightarrow 0. \end{aligned}$$

After summing over the finitely many neurons,

$$\mathcal{N}_{\omega^{(j)}, c^{(j)}} \longrightarrow \mathcal{N}_{\omega, c} \quad \text{in } H^1(D).$$

Thus the fidelity is continuous, and continuity of $z \mapsto z^q$ proves continuity of the complete objective.

Let $(\omega^{(j)}, c^{(j)})$ be a minimizing sequence. We may assume that its objective is at most the zero-network value plus one:

$$J_{\psi_k}(\omega^{(j)}, c^{(j)}) \leq \frac{1}{2} \|V\|_{H^1(D)}^2 + 1.$$

The fidelity is nonnegative, and therefore

$$\alpha \sum_{n=1}^N |c_n^{(j)}|^q \leq \frac{1}{2} \|V\|_{H^1(D)}^2 + 1.$$

In particular, all coefficients lie in a fixed interval $[-R, R]$. The sequence is consequently contained in the compact set $(\mathbb{S}^d)^N \times [-R, R]^N$. It has a convergent subsequence, and continuity of J_{ψ_k} shows that the limit is a minimizer at width N . For $N = 0$, the only admissible function is the zero function.

Step 2: a width bound. Let (ω^*, c^*) be a fixed-width minimizer with all zero coefficients discarded. The zero coefficients are feasible at the same fixed width, so

$$J_{\psi_k}(\omega^*, c^*) \leq \frac{1}{2} \|V\|_{H^1(D)}^2.$$

Since the penalty is nonnegative, this implies

$$\|\mathcal{N}_{\omega^*, c^*} - V\|_{H^1(D)} \leq \|V\|_{H^1(D)}.$$

Define

$$\bar{P}(\omega) := \langle \mathcal{N}_{\omega^*, c^*} - V, K(\omega) \rangle_{H^1(D)}.$$

Cauchy–Schwarz gives

$$\|\bar{P}\|_{C(\mathbb{S}^d)} \leq \|V\|_{H^1(D)} \sup_{\omega \in \mathbb{S}^d} \|K(\omega)\|_{H^1(D)}.$$

For every nonzero c_n^* , stationarity with respect to the n th coefficient and the preceding bound yield

$$\begin{aligned} \alpha q |c_n^*|^{q-1} &= |\bar{P}(\omega_n^*)| \\ &\leq \|V\|_{H^1(D)} \sup_{\omega \in \mathbb{S}^d} \|K(\omega)\|_{H^1(D)}. \end{aligned}$$

Since $q - 1 < 0$, reversing this power gives

$$|c_n^*| \geq \left(\frac{\alpha q}{\|V\|_{H^1(D)} \sup_{\omega \in \mathbb{S}^d} \|K(\omega)\|_{H^1(D)}} \right)^{1/(1-q)} = c_{\min}.$$

The fixed-width minimum is no larger than the zero-network value. Hence

$$N\alpha c_{\min}^q \leq \alpha \sum_{n=1}^N |c_n^*|^q \leq J_{\psi_k}(\omega^*, c^*) \leq \frac{1}{2} \|V\|_{H^1(D)}^2,$$

which gives $N \leq N_{\max}$.

Step 3: minimization over all widths. If a minimizer at a fixed width has a zero coefficient, discarding that neuron leaves the represented function and the objective unchanged. Repeating this operation gives a representation with no zero coefficients, to which Step 2 applies. Thus every fixed-width minimum is represented by a network of width at most $\lfloor N_{\max} \rfloor$. Consequently,

$$\inf_{N \in \mathbb{N}_0} \min_{(\mathbb{S}^d)^N \times \mathbb{R}^N} J_{\psi_k} = \min_{0 \leq N \leq \lfloor N_{\max} \rfloor} \min_{(\mathbb{S}^d)^N \times \mathbb{R}^N} J_{\psi_k}.$$

The right-hand side is the minimum of finitely many attained values by Step 1. This proves existence of a global minimizer and the asserted width bound. $\square$

5. ALGORITHMS

This section describes the computational pipeline: the training data are generated via the open-loop problem, candidate neurons are inserted greedily, and insertion alternates with finite-dimensional optimization of the current network.

5.1. Data generation via the open-loop problem. The optimal-control problem (P) can be written as

$$\begin{aligned} & \inf_{u, y} \mathcal{C}(u; t_0, x) \\ & \text{s.t.} \\ & E(y, u) = 0, \end{aligned}$$

with

$$E(y, u) = \begin{bmatrix} \frac{d}{dt}y - \left(f(y) + g(y)u \right) \\ y(t_0) - x \end{bmatrix}.$$

Assume that for every control the state equation admits a unique solution $y(u)$; setting $U := L^2(t_0, T; \mathbb{R}^{m_u})$, we obtain the unconstrained reduced problem

$$\inf_{u \in U} \widehat{\mathcal{C}}(u), \quad \widehat{\mathcal{C}}(u) := \mathcal{C}(u; t_0, x).$$

The reduced gradient is computed by the standard adjoint (Lagrangian) approach — see, e.g., [14] — which avoids forming the implicit derivative of $u \mapsto y(u)$: it yields

$$\nabla \widehat{\mathcal{C}}(u) = 2\eta u + g(y)^\top p \quad \text{in } U,$$

where the costate p solves the adjoint equation backward in time,

$$\begin{cases} \dot{p} + \partial_y(f(y) + g(y)u)^\top p = -\partial_y h(y), \\ p(T) = 0, \end{cases}$$

which is exactly the costate system of (TPBVP). Each gradient evaluation therefore costs one forward solve of the state equation and one backward solve of the adjoint equation. For the data-driven approach it is essential that the initial dataset $\{x^j, V(0, x^j), \nabla V(0, x^j)\}_{j=1}^M$ is *causality-free*: the samples are generated without approximate values at neighboring points. [15] provides a comprehensive survey of generation methods. As in [1], we minimize the reduced objective with Barzilai–Borwein step sizes, using the reduced gradient above; the converged costate supplies the gradient sample ∇V .

5.2. Greedy insertion. The Gâteaux derivative of (10) in a Dirac direction is

$$(42) \quad P_{\mu_t}^M(\omega) := Dl^M(\mu_t)[\delta_\omega] = \frac{1}{M} \sum_{m=1}^M ([K(\omega)](x^m) r_{\mu_t}(x^m) + \nabla K(\omega)(x^m) \cdot \nabla r_{\mu_t}(x^m)).$$

Again, the second summand is omitted for value-only fitting.

The conditional-gradient step linearizes the fidelity over a fixed total-variation ball. Its extreme points are signed Dirac measures, so the insertion search is governed by $|P_{\mu_t}^M(\omega)|$ [2]. Let $\mu_t = \mu_{\omega^{(t)}, c^{(t)}}$ be the current finite measure. For a candidate $\omega \notin \text{atom } \mu_t$, a new coefficient c , and a penalty with $\phi'(0) = 1$, the objective (47) satisfies

$$(43) \quad J^M((\omega^{(t)}, \omega), (c^{(t)}, c)) - J^M(\omega^{(t)}, c^{(t)}) = c P_{\mu_t}^M(\omega) + [\alpha + \beta w_p(\omega)]|c| + o(|c|).$$

Thus a candidate is accepted only if

$$(44) \quad |P_{\mu_t}^M(\omega)| > \alpha + \beta w_p(\omega),$$

and accepted candidates are ranked by $|P_{\mu_t}^M(\omega)| - \beta w_p(\omega)$.

For a nonhomogeneous activation, write $\omega = \lambda \hat{\omega}$ with $|\hat{\omega}| = 1$. The implementation first samples and refines directions using $|P_{\mu_t}^M(\hat{\omega})|$. For every resulting direction, it then optimizes the scalar function

$$(45) \quad \lambda \mapsto |P_{\mu_t}^M(\lambda \hat{\omega})| - \beta w_p(\lambda \hat{\omega}).$$

This two-stage search is a numerical heuristic for the joint maximization over Ω . For the activation functions used below, the finite sum in (42) satisfies $|P_{\mu_t}^M(\omega)| \leq C_t(1 + |\omega|)^{s_1}$ for a constant C_t . Hence, when $p > s_1$ and $\beta > 0$, (45) tends to $-\infty$ as $\lambda \rightarrow \infty$, by the same polynomial-growth comparison as in Corollary 3.25. The algorithm restricts λ to $\lambda \leq e^5$.

Finding a global maximizer is generally intractable; for ReLU dictionaries it is NP-hard [2]. We therefore use multiple initial directions and local L-BFGS refinement. With M value samples and Md gradient components, the convex total-variation problem underlying the conditional-gradient step has $M(d+1)$ scalar measurements and admits a minimizer with at most $M(d+1)$ atoms [2]. This bound does not assert convexity of the nonconvex coefficient problems below.

5.3. The nonconvex penalized algorithms. We use two variants of the primal–dual active point (PDAP) algorithm [25]. For Algorithm 1, the scalar penalty is

$$(46) \quad \phi_\gamma(z) := \begin{cases} \frac{z}{2} + \frac{1}{4\gamma} \log(1 + 2\gamma z), & \gamma > 0, \\ z, & \gamma = 0. \end{cases}$$

It satisfies $\phi_\gamma(0) = 0$ and $\phi'_\gamma(0) = 1$. For $\omega = (\omega_n)_{n=1}^N$ and $\mu_{\omega, c} = \sum_{n=1}^N c_n \delta_{\omega_n}$, the implemented objective is

$$(47) \quad J^M(\omega, c) := l^M(\mu_{\omega, c}) + \alpha \sum_{n=1}^N \phi_\gamma(|c_n|) + \beta \sum_{n=1}^N |c_n| w_p(\omega_n).$$

The insertion condition is (44), derived from (43). The semismooth Newton step uses the coefficient threshold $\alpha \phi'_\gamma(|c_n|) + \beta w_p(\omega_n)$.

Algorithm 2 uses $\psi_k(z) = z^q$, $q = 2/(k+1) < 1$, on the unit sphere and sets $\beta = 0$. Because $\psi'_k(0^+) = +\infty$, it uses a finite-step decrease test instead of (44).

For a fitted value function $\hat{V}$, the relative errors are

$$\begin{aligned} \text{Err}_{L^2} &:= \frac{\|\hat{V} - V\|_{L^2}}{\|V\|_{L^2}}, \\ \text{Err}_{H^1} &:= \frac{(\|\hat{V} - V\|_{L^2}^2 + \|\nabla \hat{V} - \nabla V\|_{L^2}^2)^{1/2}}{(\|V\|_{L^2}^2 + \|\nabla V\|_{L^2}^2)^{1/2}}. \end{aligned}$$

In Section 6, these norms are computed empirically on the validation or evaluation set specified for each experiment.

Algorithm 1 Iterative node insertion for J^M

- 1: Initial network $\omega^{(0)}, c^{(0)}$ of width $N(0)$.
 - 2: **while** $t < T_{\text{out}}$ **do**
 - 3: Sample N_{trial} random directions $\hat{\omega} \in \mathbb{S}^d$.
 - 4: Refine $|P_{\mu_t}^M(\hat{\omega})|$ by L-BFGS from the sampled directions and merge near-duplicate candidates.
 - 5: For each direction, optimize $\lambda > 0$ in $|P_{\mu_t}^M(\lambda\hat{\omega})| - \beta w_p(\lambda\hat{\omega})$.
 - 6: Select nodes satisfying $|P_{\mu_t}^M(\omega)| > \alpha + \beta w_p(\omega)$, ranked by $|P_{\mu_t}^M(\omega)| - \beta w_p(\omega)$, and initialize their outer coefficients by coordinate descent.
 - 7: Minimize (47) locally by the semismooth Newton method.
 - 8: Remove nodes with outer weight zero (resulting in width $N(t+1)$).
 - 9: $t = t + 1$.
 - 10: **end while**
-

For the q -penalty with $q < 1$, the zero network is always a local minimizer, and an infinitesimal insertion test gives no condition away from the existing support (Remark after Corollary 4.6). We replace it with a *finite-step decrease criterion*: given a candidate ω , we evaluate the objective decrease

$$\Delta J_{\psi_k}^M(c; \omega) = c P_{\mu_t}^M(\omega) + \frac{1}{2} c^2 \|K(\omega)\|_M^2 + \frac{\alpha}{q} |c|^q,$$

where

$$\|K(\omega)\|_M^2 := \frac{1}{M} \sum_{m=1}^M (|[K(\omega)](x^m)|^2 + |\nabla K(\omega)(x^m)|^2).$$

We accept ω if $\min_c \Delta J_{\psi_k}^M(c; \omega) < 0$. The minimizing coefficient has sign $-\text{sign}(P_{\mu_t}^M(\omega))$. Its magnitude $z^* > 0$ is the larger positive solution, computed by Newton's method, of

$$\|K(\omega)\|_M^2 z + \alpha z^{q-1} = |P_{\mu_t}^M(\omega)|,$$

and $c^* := -\text{sign}(P_{\mu_t}^M(\omega))z^*$ is used as the initial outer weight. The subsequent coefficient solve minimizes (39), whose penalty coefficient is α rather than α/q . Since $\alpha/q > \alpha$ for $q < 1$, every candidate accepted by the displayed finite-step test also strictly decreases (39). The value c^* is an initialization for that solve, not a stationary point of its objective.

Algorithm 2 Iterative node insertion for $J_{\psi_k}^M$

- 1: Initial network $\omega^{(0)}, c^{(0)}$ of width $N(0)$.
 - 2: **while** $t < T_{\text{out}}$ **do**
 - 3: Sample N_{trial} random nodes $\omega \in \mathbb{S}^d$.
 - 4: Refine $|P_{\mu_t}^M(\omega)|$ by L-BFGS from the sampled nodes and merge near-duplicate candidates.
 - 5: For each candidate, solve $\min_c \Delta J_{\psi_k}^M(c; \omega)$ and add it when $\Delta J_{\psi_k}^M(c^*; \omega) < 0$.
 - 6: Minimize (39) locally by the semismooth Newton method.
 - 7: Remove nodes with outer weight zero (resulting in width $N(t+1)$).
 - 8: $t = t + 1$.
 - 9: **end while**
-

6. NUMERICAL EXAMPLES

The examples compare the activation functions and regularizers on a smooth value function and on a value function with a discontinuous gradient. All runs use seed 42. Each reported parameter choice is run once, and the validation sets are random subsets of the PMP samples. Input normalization is computed from the full sample before the split.

6.1. Van der Pol Oscillator. We consider the optimal control of the Van der Pol oscillator:

$$\min_{u \in L^2(0,T;\mathbb{R})} \mathcal{C}(u; 0, x) = \int_0^T (y_1^2(t) + y_2^2(t) + \eta u^2(t)) dt$$

subject to

$$\begin{cases} \frac{dy_1}{dt} = y_2, \\ \frac{dy_2}{dt} = -y_1 + y_2(1 - y_1^2) + u, \\ (y_1(0), y_2(0)) = (x_1, x_2), \end{cases}$$

In this setting, $d = 2$, $m_u = 1$, $\eta = 0.1$, $T = 3$, and

$$f(y) = \begin{pmatrix} y_2 \\ -y_1 + y_2(1 - y_1^2) \end{pmatrix}, \quad g(y) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad h(y) = y_1^2 + y_2^2.$$

By the optimality condition, $u^* = -\frac{1}{2\eta} g(y^*)^\top p^* = -\frac{1}{2\eta} p_2^*$. The two-point boundary value problem (TPBVP) becomes

$$(48) \quad \begin{cases} \dot{y}_1^* = y_2^*, \\ \dot{y}_2^* = -y_1^* + y_2^*(1 - y_1^{*2}) + u^*, \\ -\dot{p}_1^* = 2y_1^* - p_2^*(1 + 2y_1^*y_2^*), \\ -\dot{p}_2^* = 2y_2^* + p_1^* + p_2^*(1 - y_1^{*2}), \\ y^*(0) = x, \quad p^*(T) = 0. \end{cases}$$

For each initial condition $x = (x_1, x_2)$ sampled on the grid $[-3, 3]^2$, we solve (48) using a Barzilai–Borwein gradient method with backtracking line search. The control is represented in a Legendre polynomial basis $u(t) = \sum_{i=0}^{N_b-1} c_i P_i(t)$ with $N_b = 30$ coefficients. The optimization yields the dataset $\{x^m, V(x^m), \nabla V(x^m)\}_{m=1}^M$, where $V(x^m) = \mathcal{C}(u^*; 0, x^m)$ and $\nabla V(x^m) = p^*(0)$.

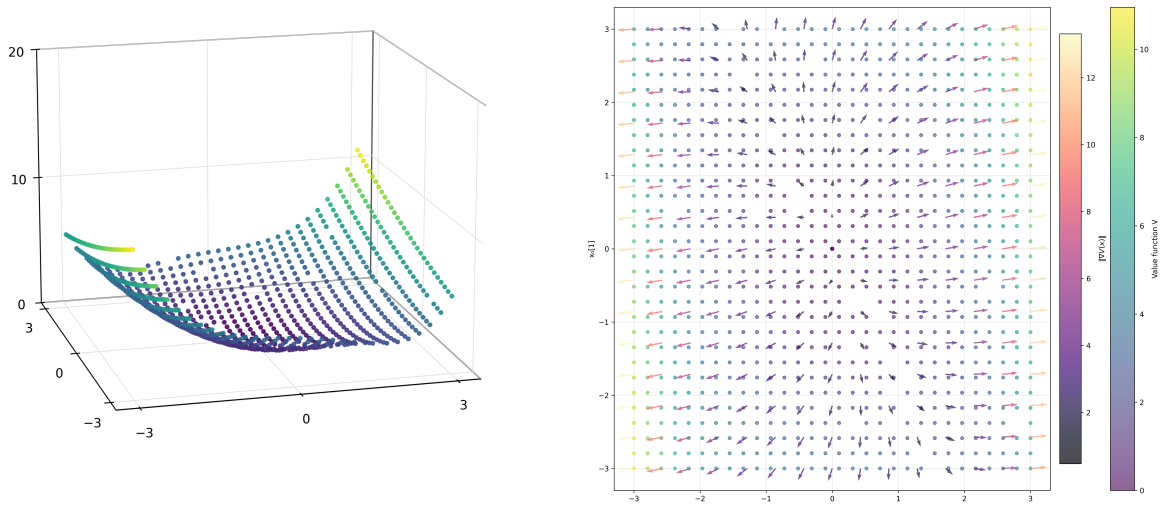


FIGURE 2. Left: value function $V(x)$ over $[-3, 3]^2$. Right: $V(x)$ (dot color) and $\nabla V(x)$ (arrows); arrow color indicates $\|\nabla V(x)\|$.

We solve the open-loop problem on a 30×30 grid over $[-3, 3]^2$ ($M = 900$ data points). All experiments run 10 outer iterations; each iteration samples $N_{\text{trial}} = 50$ candidate nodes, accepts at most 15 of them, and prunes neurons with $|c| \leq 10^{-8}$ after the local solve.

6.1.1. Effect of β and p .

Dependence on α and β . We first investigate the dependence of the validation error on the two regularization coefficients. We use Algorithm 1 with $\gamma = 1$, equal value and gradient weights, and seed 42. The tested values are

$$\alpha \in \{10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}\}, \quad \beta \in \{0, 10^{-10}, 10^{-5}, 10^{-2}, 10^{-1}\}, \quad p \in \{2.01, 2.5, 3, 4\},$$

for tanh, softplus, and the Gaussian. Since p has no effect when $\beta = 0$, the same fit is used for every displayed value of p .

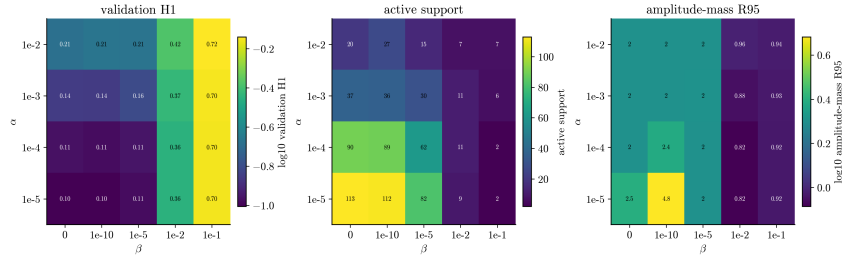


FIGURE 3. Relative H^1 validation error over the $\alpha \times \beta$ grid for the Gaussian at $p = 3$. At $\alpha = 10^{-5}$ the error changes little for $\beta \leq 10^{-5}$ and then increases; at $\alpha = 10^{-2}$ it is high for all tested values of β .

Effect of β on validation error, support size, and $R_{0.95}$. For a discrete measure $\mu = \sum_j c_j \delta_{\omega_j}$, define

$$R_{0.95}(\mu) := \inf \{R \geq 0 : |\mu|(B_R) \geq 0.95 |\mu|(\Omega)\}.$$

For the Gaussian with $p = 3$ and $\alpha = 10^{-5}$, moving from $\beta = 0$ to $\beta = 10^{-5}$ changes the validation H^1 error from 0.0994 to 0.1053 while the support falls from 113 to 82 neurons and $R_{0.95}$ from 2.49 to 2.01. At $\beta = 10^{-2}$, the error is 0.3556, the support contains 9 neurons, and $R_{0.95} = 0.82$. The smallest error among the runs with $\beta > 0$ is 0.0989, obtained for the Gaussian at $\alpha = 10^{-5}$, $\beta = 10^{-10}$, and $p = 3$.

For tanh and $\beta = 10^{-1}$, the algorithm returns the zero measure for every tested value of α and p . At $\mu = 0$, no candidate satisfies the insertion condition.

Effect of p on $R_{0.95}$. Lemma 3.21 gives

$$R_* = \max \left\{ 1, \left(\frac{2^{s_1} C_P}{\beta} \right)^{1/(p-s_1)} \right\}.$$

For softplus ($s_1 = 1$), the β -dependent factor $\beta^{-1/(p-s_1)}$ is of order 10^{10} at $p = 2.01$, $\beta = 10^{-10}$, but only of order 10^3 at $p = 4$ and the same β . The unknown residual-dependent factor $2^{s_1} C_P$ prevents an absolute value of R_* from being inferred from these experiments. For softplus with $\alpha = 10^{-4}$ and $\beta = 10^{-2}$, the observed $R_{0.95}$ values are 1.74, 1.43, and 1.09 for $p = 2.5, 3$, and 4, respectively. This observed decrease is consistent with the dependence of R_* on p , although C_P differs between the fitted networks.

Upper bound on the radial parameter. The algorithm restricts λ to $\lambda \leq e^5$. For the Gaussian runs reported above, $R_{0.95} \leq 2.49 \ll e^5$; hence at least 95% of the total variation lies well inside this upper bound.

6.1.2. Algorithm 1: log penalty and moment norm. We investigate the effect of including gradient data for softplus, tanh, and the Gaussian. Table 1 compares fitting the values alone with fitting both values and gradients.

TABLE 1. Effect of gradient-augmented fitting ($\gamma = 1$, $\alpha = 10^{-5}$, $\beta = 10^{-10}$; $p = 2.01$ for softplus and tanh, and $p = 3$ for the Gaussian).

Activation	L^2 trained	H^1 trained	Activation	L^2 trained	H^1 trained
softplus	8.70×10^{-2}	4.31×10^{-1}	softplus	5.63×10^{-1}	1.25×10^{-1}
tanh	4.46×10^{-2}	4.43×10^{-1}	tanh	4.90×10^{-1}	1.67×10^{-1}
Gaussian	3.28×10^{-2}	4.15×10^{-1}	Gaussian	4.71×10^{-1}	9.89×10^{-2}

(A) Effect of gradient training on Err_{L^2} (B) Effect of gradient training on Err_{H^1}

Choice of activation function in gradient training. We compare softplus, tanh, and the Gaussian. Because the gradient-augmented loss contains both the value and gradient errors, $K(\omega)$ contributes to the approximation of V and $\nabla K(\omega)$ contributes to the approximation of ∇V . Figure 4 contrasts the three activations through their value, first derivative, and second derivative. The softplus derivative increases from 0 to 1, the derivative of tanh tends to zero at both ends of the real line, and the Gaussian derivative is localized and changes sign. These derivatives generate different columns $a\rho'(a \cdot x + b)$ in the gradient fitting problem.

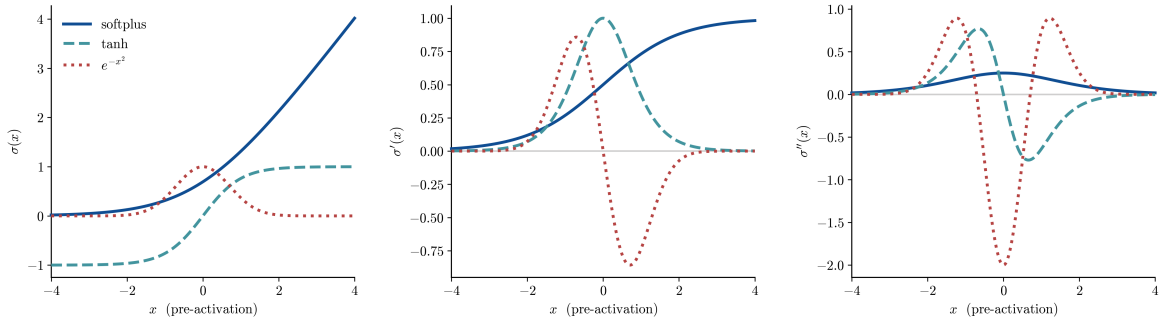


FIGURE 4. The three smooth profiles used by Algorithm 1 — softplus, tanh, and the Gaussian e^{-z^2} — shown as ρ , ρ' , and ρ'' . Under the gradient-augmented loss, the first derivative is the basis that reconstructs ∇V .

For the tested parameters, the Gaussian has the smallest relative H^1 error. All three approximate the smooth value function, with different support sizes.

TABLE 2. Support size and validation error for the parameter values in Table 1.

Activation	N	Err_{L^2}	Activation	N	Err_{H^1}
softplus	62	8.70×10^{-2}	softplus	35	1.25×10^{-1}
tanh	109	4.46×10^{-2}	tanh	76	1.67×10^{-1}
Gaussian	97	3.28×10^{-2}	Gaussian	112	9.89×10^{-2}

(A) L^2 trained (B) H^1 trained

With the H^1 loss, all three gradient errors fall substantially. The Gaussian is most accurate (9.89×10^{-2}) and uses 112 atoms. Softplus reaches 1.25×10^{-1} with only 35 atoms, while tanh reaches 1.67×10^{-1} with 76.

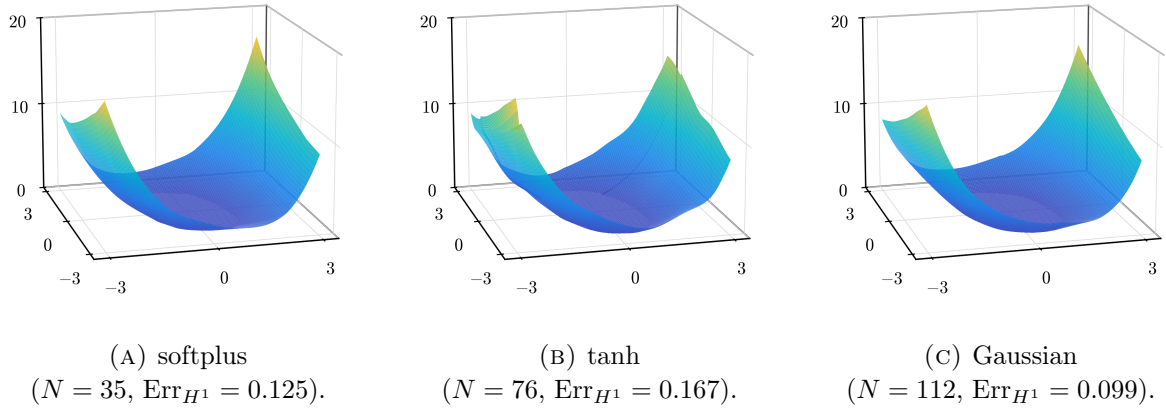


FIGURE 5. Learned value functions under the H^1 loss using the parameter values in Table 1. All three approximate the smooth value surface; they differ in support size and accuracy.

6.1.3. *Algorithm 2: $|c|^q$ -penalty with ReLU^k activation.* Algorithm 2 uses the profile $\rho(z) = \text{ReLU}(z)^k$ with the penalty ψ_k , namely $\alpha \sum_i |c_i|^q$ with $q = 2/(k+1)$. The swept parameter is the power k ; all fits below use $\alpha = 10^{-5}$.

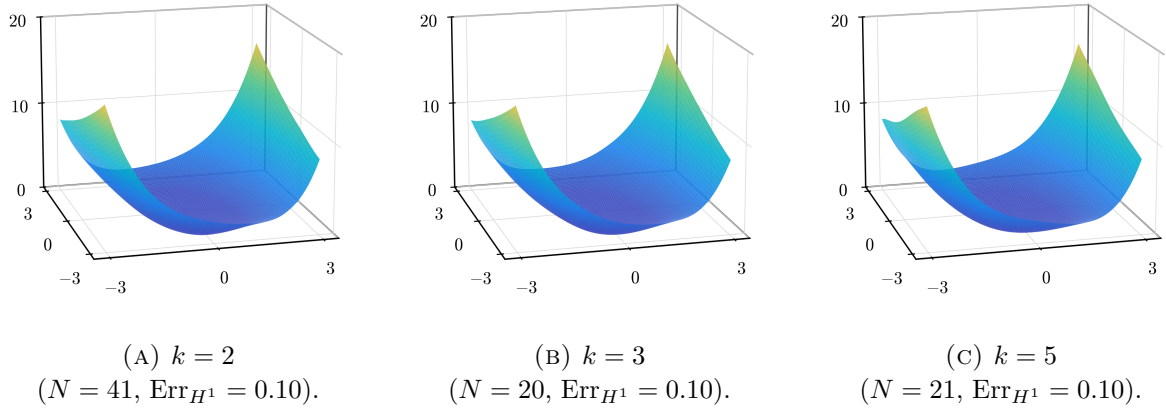


FIGURE 6. H^1 -trained networks with $\alpha = 10^{-5}$ for $k \in \{2, 3, 5\}$.

Under the L^2 loss the gradient error stays large for every power ($\text{Err}_{H^1} \approx 4\text{--}5 \times 10^{-1}$): the value-only fit ignores ∇V , so the atoms are placed without regard to the gradient field. The H^1 loss drives the gradient error down to $\approx 1.0 \times 10^{-1}$ for *all* powers (Table 3).

Increasing k changes both the activation and the induced exponent $q = 2/(k+1)$. The support drops from 41–42 atoms for $k = 2$ and 2.01 to 20–22 atoms for $k = 3, 4, 5$, while the error remains between 0.098 and 0.104. Under the L^2 loss the fits use 10–14 atoms and have $\text{Err}_{H^1} \approx 4\text{--}5 \times 10^{-1}$ (Table 3(a)).

TABLE 3. Algorithm 2: network size and validation error vs. the atom power k ($\alpha = 10^{-5}$).

Power	N	Err_{L^2}	Err_{H^1}	Power	N	Err_{L^2}	Err_{H^1}
$k = 2$	14	5.05×10^{-2}	4.83×10^{-1}	$k = 2$	41	4.16×10^{-1}	9.84×10^{-2}
$k = 2.01$	14	4.83×10^{-2}	4.81×10^{-1}	$k = 2.01$	42	4.17×10^{-1}	9.93×10^{-2}
$k = 3$	12	5.98×10^{-2}	4.94×10^{-1}	$k = 3$	20	4.17×10^{-1}	1.00×10^{-1}
$k = 4$	12	5.48×10^{-2}	4.61×10^{-1}	$k = 4$	22	4.16×10^{-1}	9.97×10^{-2}
$k = 5$	10	4.77×10^{-2}	4.04×10^{-1}	$k = 5$	21	4.14×10^{-1}	1.04×10^{-1}

(A) L^2 trained

(B) H^1 trained

6.1.4. *Summary.* Figure 7 plots the running best relative H^1 error. The Gaussian reaches 9.89×10^{-2} with 112 atoms, while ReLU² and ReLU⁵ reach comparable errors with roughly 40 and 20 atoms. Softplus stops at 1.25×10^{-1} with 35 atoms, and therefore uses fewer atoms than the Gaussian at a larger error.

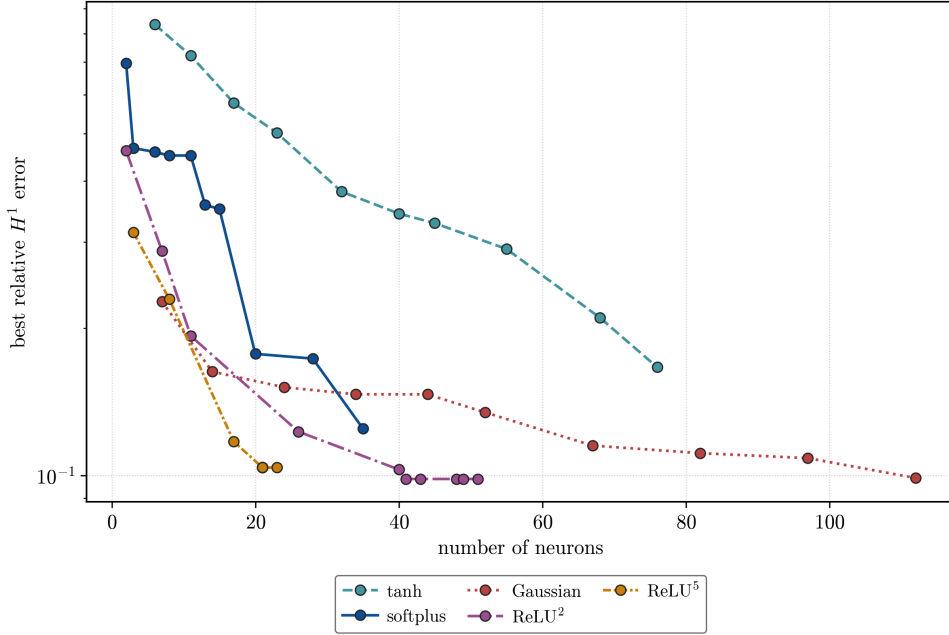


FIGURE 7. Running best relative H^1 validation error against support size. Non-homogeneous profiles use the parameter values stated in Table 1; homogeneous profiles use $\alpha\psi_k$ at $\alpha = 10^{-5}$.

We synthesize feedback laws from $y_0 = (2, 1)$ and roll them out over the horizon $T = 12$ with fixed-step RK4, zero-order-hold controls clipped to $|u| \leq 50$ as a numerical guard. The reference law interpolates the time-zero costate component of the dataset with a C^1 Clough–Tocher interpolant and applies it as a stationary feedback. The finite-horizon optimal feedback is time-dependent and the rollout horizon exceeds the data horizon $T = 3$. Therefore its rollout cost 6.48 is not $V(0, y_0)$, and the comparison is not a test of finite-horizon optimality. Figure 8 shows $\|y(t)\|$ and $|u(t)|$ for softplus, the Gaussian, and ReLU^k ($k = 5$), beside this reference. All three drive the state to the origin and track the reference control, at essentially its closed-loop cost (6.63, 6.58, 6.49 vs. 6.48).

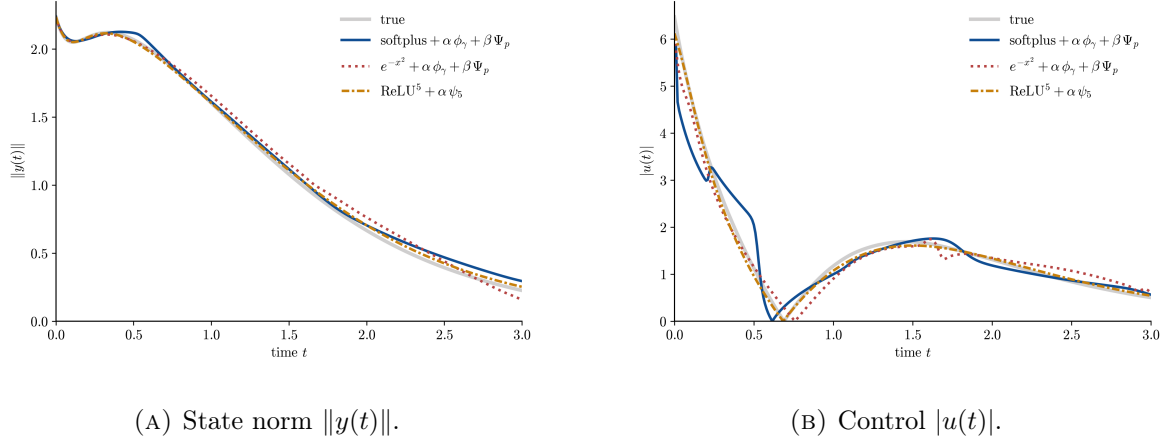


FIGURE 8. Evolution of the state and control from $y_0 = (2, 1)$ under the feedback synthesized from each fitted value function and under the stationary reference law obtained from the time-zero costates.

Figure 9 shows the inner parameters for the Gaussian, softplus, and ReLU^k ($k = 5$). The two algorithms produce structurally different representations: Algorithm 2 constrains its atoms to the unit sphere $\mathbb{S}^2$, while Algorithm 1 leaves them free — the Gaussian atoms in particular span a wide range of norms.

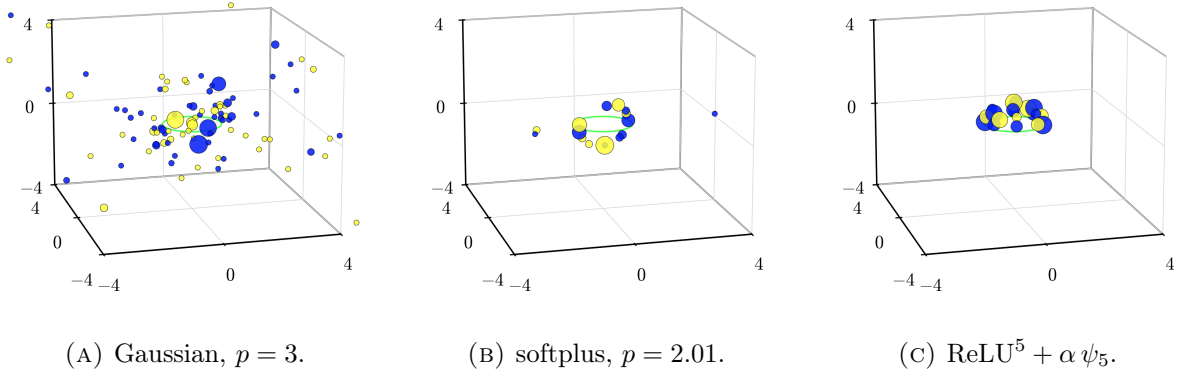


FIGURE 9. Inner parameters (a_1, a_2, b) , with the unit circle in the plane $b = 0$ shown in light green. The two nonhomogeneous fits use $\alpha = 10^{-5}$, $\beta = 10^{-10}$, $\gamma = 1$; dot color gives the sign and dot size the magnitude of the outer weight.

6.2. Pendulum Swing-Up. The question in this example is how a sparse gradient-trained network approximates a continuous value function whose gradient jumps across a switching set. A finite network with a continuously differentiable activation has a continuous gradient and therefore cannot reproduce the jump exactly. Rectified homogeneous activations can place changes of slope along hyperplanes. We compare these two classes at the support sizes returned by Algorithms 1 and 2.

Figure 10 contrasts three activation functions. The rectified linear unit ReLU has a step derivative, so a single neuron can represent a gradient jump on its hyperplane. The derivative of ReLU^2 is continuous and piecewise linear, whereas the softplus derivative is smooth. The positive homogeneity of the rectified functions allows the fractional-exponent formulation of Section 4.

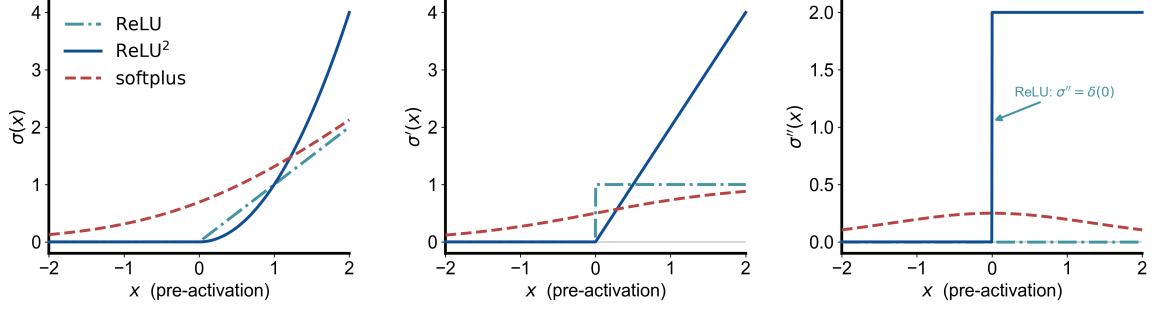


FIGURE 10. The profiles ρ , ρ' , and ρ'' for ReLU, ReLU², and softplus. The derivative of ReLU jumps, the derivative of ReLU² has a kink, and the softplus derivative is smooth. The second distributional derivative of ReLU is a Dirac measure at the origin.

The example we consider follows from [12]. We write the state as

$$x = (x_1, x_2)^\top = (\theta, \dot{\theta})^\top,$$

the pendulum angle and angular velocity. (The letter ω remains reserved for the network parameters.) We consider the optimal control problem

$$\min_{u \in L^2(0, \infty; \mathbb{R})} \mathcal{C}(u; 0, x) = \int_0^\infty (\sin^2 \theta(t) + (\cos \theta(t) - 1)^2 + \dot{\theta}^2(t) + \eta u^2(t)) dt$$

subject to

$$\begin{cases} m\ell^2 \ddot{\theta} = -b\dot{\theta} + mg\ell \sin \theta + u, \\ (\theta(0), \dot{\theta}(0)) = (x_1, x_2). \end{cases}$$

The target equilibria are $x_{U,j} := (2j\pi, 0)$, $j \in \mathbb{Z}$; for a selected target we write $x_U = x_{U,j}$.

In this setting the state dimension is $d = 2$ and the control is scalar. The physical and cost parameters are $m = \ell = 1$, $b = 0.1$, $g = 9.8$, and $\eta = 1$. To initialize the backward integration, we use the local linear-quadratic regulator (LQR) approximation near x_U . Let $\tilde{x} = x - x_U$ be the local coordinate around the upright equilibrium. Linearizing the dynamics at $(x_U, 0)$ gives

$$\dot{\tilde{x}} = A\tilde{x} + Bu, \quad A = \begin{pmatrix} 0 & 1 \\ \frac{g}{\ell} & -\frac{b}{m\ell^2} \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ \frac{1}{m\ell^2} \end{pmatrix}.$$

The running cost is locally approximated by

$$|\tilde{x}|^2 + \eta u^2.$$

The local infinite-horizon LQR value function is

$$V^\infty(\tilde{x}) = \tilde{x}^\top P \tilde{x},$$

where P is the positive definite solution of the algebraic Riccati equation

$$A^\top P + PA - \frac{1}{\eta} P B B^\top P + I_2 = 0.$$

The superscript ∞ refers to the infinite-horizon LQR problem. For $\varepsilon > 0$, we define the local LQR sublevel set and boundary by

$$\mathcal{E}_\varepsilon = \{\tilde{x} : V^\infty(\tilde{x}) \leq \varepsilon\}, \quad \partial \mathcal{E}_\varepsilon = \{\tilde{x} : V^\infty(\tilde{x}) = \varepsilon\}.$$

The corresponding state-costate system becomes

$$(49) \quad \begin{cases} m\ell^2 \ddot{\theta}^* = -b\dot{\theta}^* + mg\ell \sin \theta^* + u^*, \\ -\dot{p}_1^* = 2 \sin \theta^* + \frac{g}{\ell} \cos \theta^* p_2^*, \\ -\dot{p}_2^* = 2\dot{\theta}^* + p_1^* - \frac{b}{m\ell^2} p_2^*, \\ x^*(0) = x, \quad x^*(T_x) - x_U \in \partial\mathcal{E}_\varepsilon, \quad p^*(T_x) = \nabla V^\infty(x^*(T_x) - x_U). \end{cases}$$

In the experiment we use $\varepsilon = 2 \times 10^{-4}$, and on $\partial\mathcal{E}_\varepsilon$ the terminal costate is

$$p^*(T_x) = \nabla V^\infty(\tilde{x}^*(T_x)) = 2P\tilde{x}^*(T_x).$$

For the data generation, we integrate 2000 PMP paths backward from the local LQR boundary $\partial\mathcal{E}_\varepsilon$. Along each path, the accumulated running cost plus the boundary value ε gives $V(x)$, and the costate gives $\nabla V(x) = p$. The backward integration stops when the accumulated value reaches 100; the training samples are restricted to values below 50. This yields the dataset

$$\{x^m, V(x^m), \nabla V(x^m)\}_{m=1}^M.$$

The construction provides samples on both sides of the gradient discontinuity. For $j \in \mathbb{Z}$, let $V_{2j\pi}(x)$ be the fixed-target value obtained by minimizing over paths that reach the local LQR neighbourhood of $(2j\pi, 0)$. The stationary value is

$$V(x) = \min_{j \in \mathbb{Z}} V_{2j\pi}(x).$$

In the region where the targets $(0, 0)$ and $(2\pi, 0)$ are minimizing, this reduces to $V(x) = \min\{V_0(x), V_{2\pi}(x)\}$. Define the representative switching set between these adjacent targets by

$$S_{0,2\pi} := \{x \in \mathbb{R}^2 : V_0(x) = V_{2\pi}(x) = V(x)\}.$$

On the numerical domain $D = [-8, 8]^2$, the periodic switching set and its closed δ -neighbourhood are

$$S := D \cap \bigcup_{j \in \mathbb{Z}} (S_{0,2\pi} + (2j\pi, 0)), \quad S_\delta := \{x \in D : \text{dist}(x, S) \leq \delta\}, \quad \delta > 0.$$

The regional errors below use $\delta = 0.3$. At points of S , two adjacent fixed-target values agree while their gradients may differ; the sampled PMP costates provide the corresponding one-sided gradients.

The generator locates $S_{0,2\pi}$ by intersecting equal-value contours with their 2π -shifted copies and uses periodic translates in D . To obtain samples on both sides of S , the raw paths are shifted by $\pm 2\pi$ in θ . A shifted point within distance 0.5 of S is retained when its value agrees with the corresponding fixed-target value V_0 or $V_{2\pi}$ and is at least 0.3 below the first-order extrapolation of the other value. The latter is computed from the 32 nearest raw samples within radius 0.1. This adds 900 samples, 300 on one side of S and 600 on the other. The final dataset contains 3,900 samples: 3,000 from the backward PMP paths and 900 near S . It therefore contains gradient samples from both sides of the discontinuity. Figure 11 shows the dataset generated.

For 23 tested parameter choices, at least 95% of the total variation is carried by atoms at the upper radial bound $\lambda = e^5$. These runs are retained in the numerical record but excluded when selecting the nonhomogeneous networks reported below.

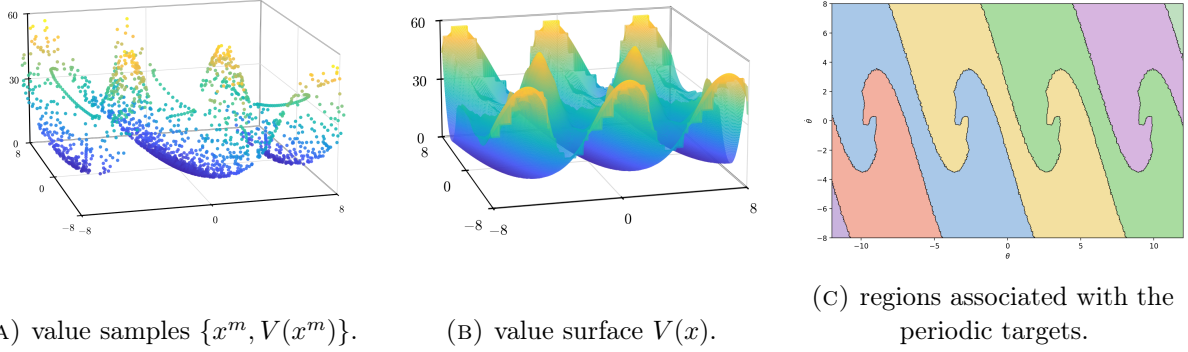


FIGURE 11. Optimal value function $V(x)$ over the $(\theta, \dot{\theta})$ plane, 2π -periodic in θ . (a) The raw (x, V) samples produced by the PMP generator. (b) The reconstructed value surface. (c) Regions where V_0 or another periodic fixed-target value is minimal; their boundaries form S , across which ∇V may be discontinuous.

6.2.1. *Approximation near the switching set.* Figure 12 shows the fitted surfaces on the same physical window and value range as the value function in Figure 11b. The cross-section below compares the approximations near S .

Let x_S be the selected regular point of S in the region with the highest sample density, and let n be a unit normal to S at x_S . Figure 13 shows the cross-section $x(s) = x_S + sn$. Along this section, $V = \min\{V_0, V_{2\pi}\}$. It has a kink at $s = 0$, and the normal gradient $n \cdot \nabla V$ jumps there. The training set contains samples on both sides of this point. No fitted model reproduces the magnitude of the jump: all undershoot the steep gradient on the side $s < 0$. ReLU² has the smallest error on $S_{0.3}$ and develops the sharpest change of slope near $s = 0$; the smooth Gaussian, softplus, and tanh fits interpolate through the discontinuity. The higher rectified powers remain smooth enough in their first derivative to miss the jump at the fitted budgets.

We next ask whether adding samples near S reduces the error on $S_{0.3}$. We compare 6,000-sample sets containing 23%, 40%, or 60% of their samples near S with a set obtained by adding 2,000 such samples to the first set. The Gaussian and ReLU² networks are evaluated on the same disjoint set of approximately 9.4×10^5 points.

For the tested sets, increasing the proportion of samples near S does not reduce the error on $S_{0.3}$ (Table 4). Thus the observed difficulty is not resolved by additional local samples.

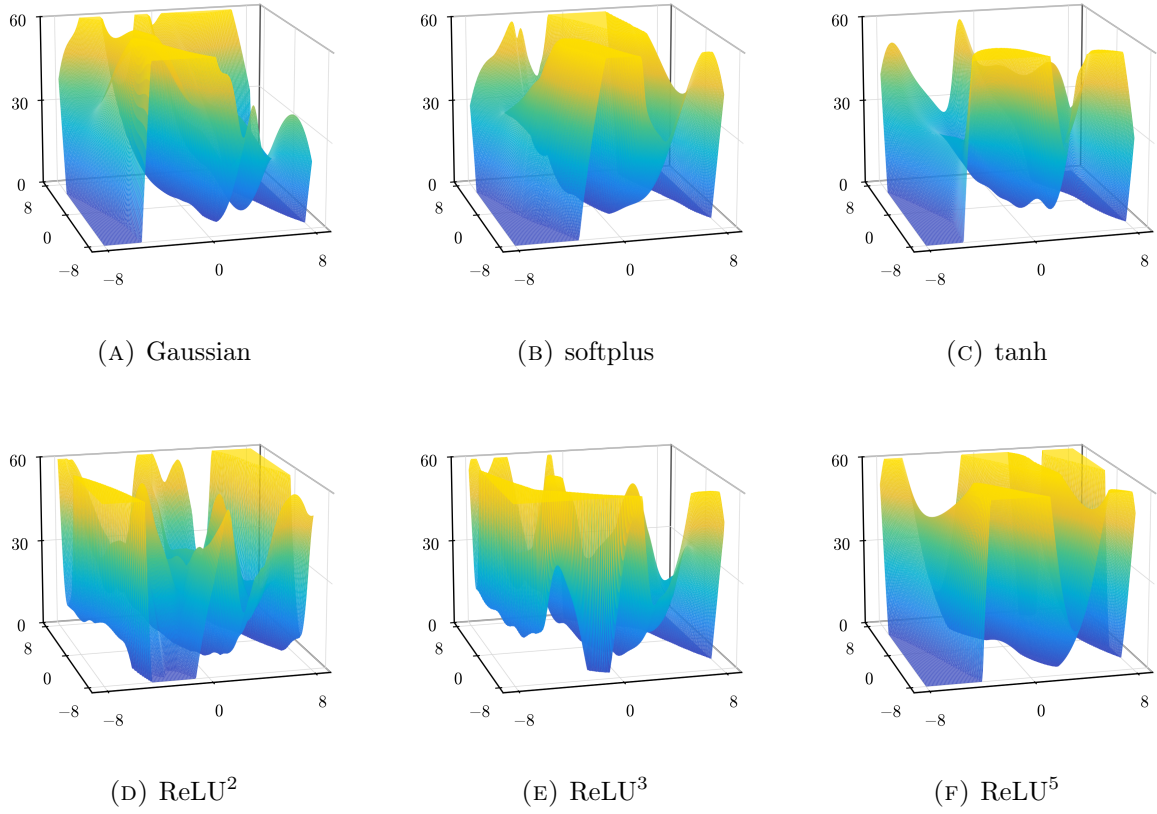


FIGURE 12. Learned value surfaces $\widehat{V}(x)$ over the $(\theta, \dot{\theta})$ window $[-8, 8]^2$, displayed on the common range $\widehat{V} \in [0, 60]$. The ReLU^k models are trained with the fractional-exponent penalty ψ_k , and the nonhomogeneous models with (47) using the parameter values reported in Table 5.

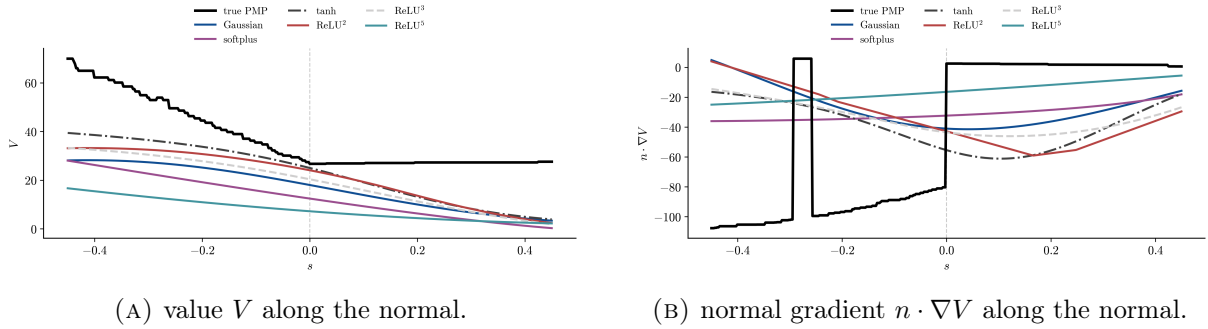


FIGURE 13. Normal cross-section of the switching curve, parametrized by the normal distance s . Left: the true value (dashed) is $\min\{V_0, V_{2\pi}\}$ on this cross-section and has a kink at $s = 0$. Right: the true normal gradient jumps at $s = 0$; ReLU^2 develops the sharpest fitted change of slope, while the smooth models interpolate through the discontinuity. The rectangular excursion of the reference curve near $s \approx -0.3$ is a nearest-neighbor artifact in the evaluation of $\min\{V_0, V_{2\pi}\}$, not a feature of V .

Model	Training set	$S_{0.3}$	$D \setminus S_{0.3}$	N
Gaussian	6k, 23% near S	6.25×10^{-1}	6.12×10^{-1}	69
	6k, 40% near S	6.05×10^{-1}	6.02×10^{-1}	106
	6k, 60% near S	6.17×10^{-1}	6.99×10^{-1}	91
	6k + 2k near S	6.17×10^{-1}	6.53×10^{-1}	88
ReLU ²	6k, 23% near S	2.46×10^{-1}	1.56×10^{-1}	108
	6k, 40% near S	2.89×10^{-1}	1.72×10^{-1}	131
	6k, 60% near S	3.46×10^{-1}	2.35×10^{-1}	109
	6k + 2k near S	2.88×10^{-1}	1.84×10^{-1}	131

TABLE 4. Relative H^1 error on $S_{0.3}$ and $D \setminus S_{0.3}$. For each model and training set, the table reports the result with the smallest error on $S_{0.3}$ among the three tested values of α , with every entry in that row taken from the same run. The Gaussian uses $(\beta, p, \gamma) = (10^{-4}, 2.01, 0)$ and $\alpha \in \{10^{-3}, 10^{-4}, 10^{-5}\}$; ReLU² uses ψ_2 , $\beta = 0$, and $\alpha \in \{10^{-4}, 10^{-5}, 10^{-6}\}$. The symbol N denotes the support size.

6.2.2. ReLU² fits the discontinuous gradient best. Among the tested activation functions, ReLU² has the smallest error in the value and gradient near S .

Figure 14 tracks the best relative H^1 validation error reached as neurons are inserted, for each activation paired with its penalty. The three nonhomogeneous activations use the parameter values in Table 5; the rectified powers use ψ_k , with $\alpha = 10^{-6}$ for $k \in \{2, 3\}$ and $\alpha = 10^{-4}$ for $k = 5$. The error for ReLU² becomes smaller than the other errors within the first insertions and reaches $\approx 3.1 \times 10^{-1}$. ReLU³ is second, while softplus, tanh, and the Gaussian remain above 5×10^{-1} and ReLU⁵ has the largest final error. Among the rectified powers, the error increases with k ; ReLU³ remains second overall, while the advantage is lost by $k = 5$.

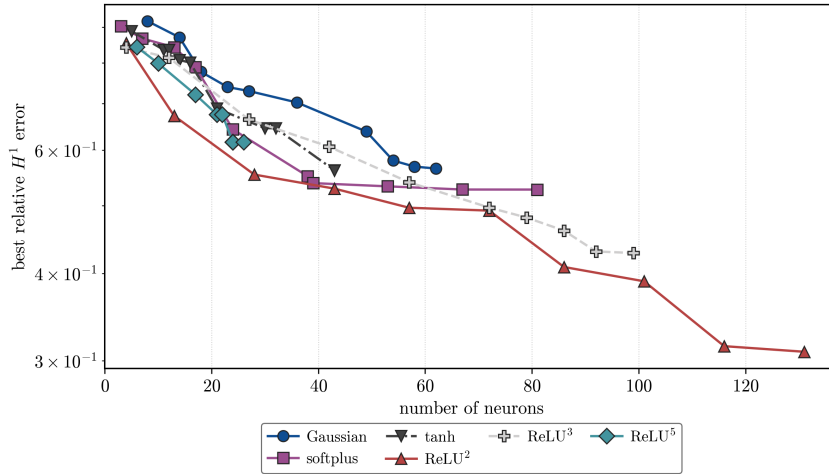


FIGURE 14. Best relative H^1 validation error against neuron count. The non-homogeneous activations use (47); the homogeneous activations use $J_{\psi_k}^M$ with $\alpha = 10^{-6}$ for $k \in \{2, 3\}$ and $\alpha = 10^{-4}$ for $k = 5$.

The same ordering holds region by region. Figure 15 reports the relative H^1 error on a separate evaluation set of approximately 9.6×10^5 points. Training rows are excluded, and for each run we use the recorded iterate with the smallest training objective. Errors are reported on

$S_{0.3}$ and $D \setminus S_{0.3}$. ReLU² has the smallest error in both sets, followed by ReLU³. Among the moment-regularized profiles, softplus has the smallest errors, followed by tanh and the Gaussian. ReLU⁵ is the one model whose two regional errors are nearly identical and largest.

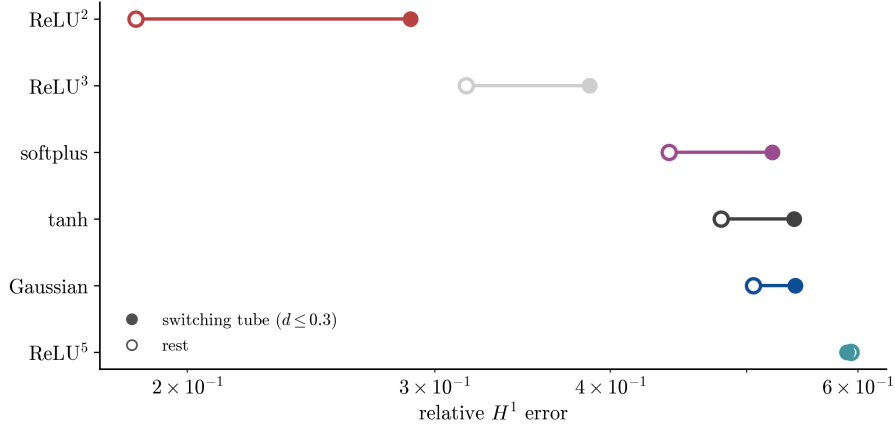


FIGURE 15. Relative H^1 error on $S_{0.3}$ (filled markers) and $D \setminus S_{0.3}$ (open markers), with rows ordered by the latter error.

Figure 16 draws the line $\{a \cdot x + b = 0\}$ associated with each atom in the physical $(\theta, \dot{\theta})$ plane, with opacity proportional to the outer weight $|c_n|$, for ReLU² (left) and Gaussian (right); the switching curve is shown in black. The lines with the largest coefficients for ReLU² are approximately parallel to the diagonal parts of S . The Gaussian lines have a wider range of orientations. Each Gaussian atom $e^{-(a \cdot x + b)^2}$ is constant along its line.

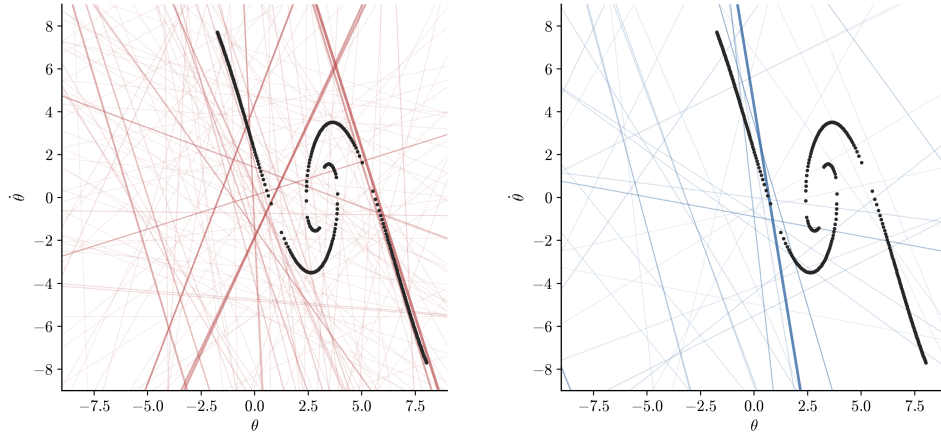


FIGURE 16. Lines $\{a \cdot x + b = 0\}$ in the $(\theta, \dot{\theta})$ plane (opacity $\propto |c_n|$) for ReLU² (left) and Gaussian (right); the switching curve is black. ReLU²'s strongest ridges run approximately parallel to the switching set; the Gaussian atoms have a wider range of orientations.

6.2.3. *Feedback synthesis across the switching set.* Minimizing the Hamiltonian in u gives the optimal control in feedback form,

$$u(x) = -\frac{1}{2\eta m\ell^2} \partial_{\dot{\theta}} V(x);$$

replacing V by a fitted $\hat{V}$ yields a candidate feedback law. We test each law in closed loop from two initial states placed on either side of the switching curve, $A = (0.71, 0.68)$ and $B = (0.23, 0.53)$, integrating the dynamics over the horizon $T = 10$ with fixed-step RK4 ($\Delta t = 5 \times 10^{-3}$), zero-order-hold controls, and the numerical bound $|u| \leq 30$ (the OCP itself is unconstrained). The optimal paths from the two starts are different. From A , the optimal law swings the pendulum over the top to the upright at $\theta = 2\pi$, while from B it brakes directly to the upright at $\theta = 0$. Thus it must select the path associated with $V_{2\pi}$ at A and the path associated with V_0 at B . The reference law uses the 40 nearest raw PMP samples to compare V_0 and $V_{2\pi}$ and applies the feedback formula using the costate associated with the smaller value.

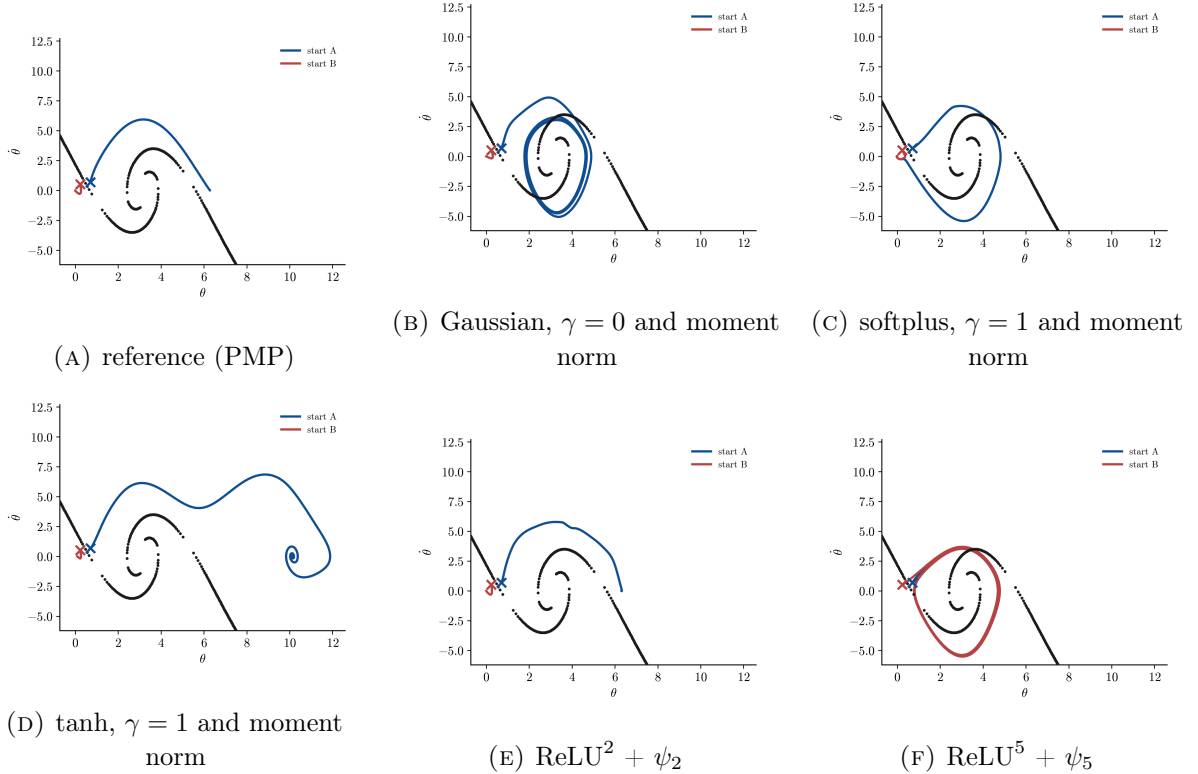


FIGURE 17. Closed-loop paths in the $(\theta, \dot{\theta})$ plane from the two starts A (blue) and B (red) on either side of the switching curve (black), one panel per feedback law; all panels share the same axes. The reference law swings over the top from A and brakes to $\theta = 0$ from B ; ReLU^2 reproduces both behaviors.

Two laws reach the upright neighbourhood from both starts, ReLU^2 and softplus, and they do so at very different costs (Table 5). From B , ReLU^2 brakes to the upright at essentially the reference cost (10.3 vs 10.2); from A it follows the path associated with $V_{2\pi}$, with cost 57.9 rather than the reference cost 26.2. Softplus reaches the upright from both starts as well, but at 89.2 and 28.2, three times the reference from A and nearly three times from B . The Gaussian, tanh, and ReLU^3 satisfy the terminal criterion only from B , while ReLU^5 satisfies it from neither start. Figure 18 shows the control signal from B : the ReLU^2 law tracks the reference almost exactly, while the others oscillate or saturate.

Effect of β with α , γ , and p fixed. The parameter values in Table 5 differ in α and γ as well as in β . To vary only β , we fix softplus at $\alpha = 10^{-5}$, $\gamma = 10$, $p = 2.01$ with the H^1 loss and vary only β (Table 6). The $\beta = 0$ fit has 77 neurons and relative H^1 error 0.515.

Feedback law	start $A = (0.71, 0.68)$		start $B = (0.23, 0.53)$	
	cost	reaches upright	cost	reaches upright
reference (PMP)	26.2	yes	10.2	yes
Gaussian	186.0	no	15.2	yes
softplus	89.2	yes	28.2	yes
tanh	434.5	no	18.9	yes
$\text{ReLU}^2 + \psi_2$	57.9	yes	10.3	yes
$\text{ReLU}^3 + \psi_3$	29261.8	no	10.3	yes
$\text{ReLU}^5 + \psi_5$	235.9	no	232.4	no

TABLE 5. Closed-loop rollout cost over the horizon $T = 10$ and terminal outcome for each feedback law from the two starts on either side of the switching curve; “reaches upright” means the final state satisfies $\text{dist}(\theta, 2\pi\mathbb{Z}) < 0.4$ and $|\dot{\theta}| < 0.4$ at $T = 10$. This condition concerns the terminal state of one finite rollout and does not assert asymptotic stability. The reference law compares V_0 and $V_{2\pi}$ using the nearest PMP samples. The middle block uses Algorithm 1 (Gaussian $\alpha = 10^{-4}$, $\beta = 10^{-4}$, $p = 2.01$, $\gamma = 0$; softplus $\alpha = 10^{-4}$, $\beta = 10^{-10}$, $p = 2.01$, $\gamma = 1$; tanh $\alpha = 10^{-4}$, $\beta = 10^{-5}$, $p = 3$, $\gamma = 1$), the lower block uses Algorithm 2 at $\beta = 0$, with $\alpha = 10^{-6}$ for $k \in \{2, 3\}$ and $\alpha = 10^{-4}$ for $k = 5$.

β	N	Err_{H^1}	$R_{0.95}$	cost A	cost B
0	77	0.515	1.00	73013.3	66996.7
10^{-10}	69	0.530	1.00	168.3	170.2
10^{-5}	34	0.711	1.00	87.5	92.2
10^{-2}	7	0.857	2.20	82.6	86.0

TABLE 6. Softplus with fixed parameters ($\alpha = 10^{-5}$, $\gamma = 10$, $p = 2.01$, seed 42), varying only β . No row reaches the upright from either start.

Every tested value $\beta > 0$ reduces the rollout costs relative to $\beta = 0$; at $\beta = 10^{-2}$ they are 82.6 and 86.0. None of these four networks reaches the upright from either start. The softplus network in Table 5 also uses different values of α and γ , so its result cannot be attributed to β alone.

In summary, ReLU^2 with the fractional-exponent penalty gives the lowest error for this nonsmooth target and the rollout closest to the reference: it reaches the upright from both starts, at cost 10.3 from B and 57.9 from A . The softplus network in Table 5 also reaches the upright from both starts, at the larger costs 28.2 and 89.2. The Gaussian, tanh, and ReLU^3 succeed only from B , while ReLU^5 succeeds from neither. Adding samples near S does not reverse this ordering of the validation errors.

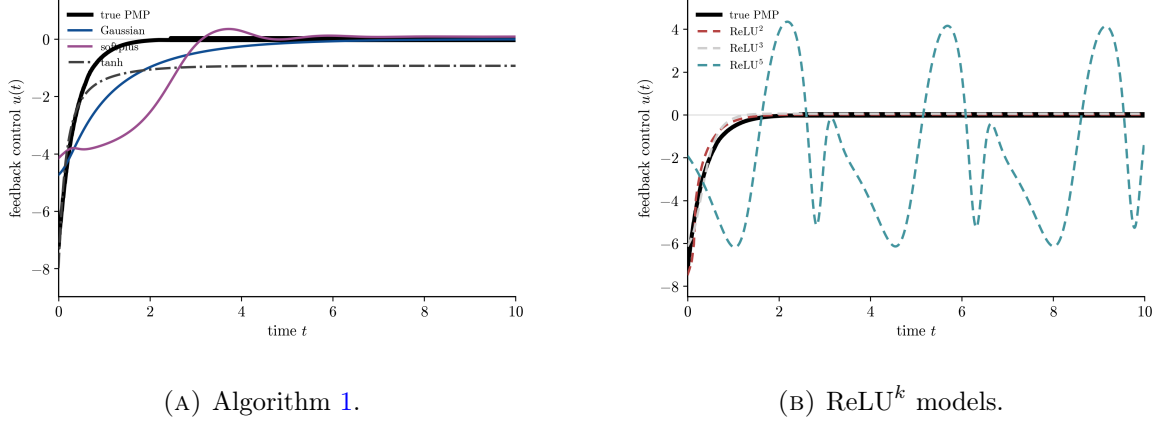


FIGURE 18. Feedback control $u(t)$ from start B , per law, beside the reference (black). The vertical axis is truncated; the softplus control reaches the numerical bound ± 30 . The reference brakes to $\theta = 0$ with u rising from ≈ -7 to 0; ReLU^2 tracks it almost exactly.

7. CONCLUSION AND OUTLOOK

We developed sparse gradient-augmented neural-network approximations of value functions from PMP data on an unbounded parameter space. Two activation classes require different treatments of the radial parameter. For nonhomogeneous profiles, the objective

$$J(\mu) = L(\mu) + \alpha \Phi_\phi(\mu) + \beta \|\mu\|_{\mathcal{M}_p(\Omega)}$$

combines the concave functional Φ_ϕ with the weighted total variation $\|\mu\|_{\mathcal{M}_p(\Omega)}$. For positively k -homogeneous profiles, radial scaling is absorbed into the outer coefficient, and the equivalent problem on the unit sphere has the fractional-power penalty $|c|^{2/(k+1)}$.

For a nonhomogeneous profile of polynomial growth $s_0 < p$, the direct method of the calculus of variations using narrow convergence gives a global minimizer (Theorem 3.16). If its full H^1 atom has growth order $s_1 < p$, then

$$|\bar{P}(\omega)| \leq C_K \|r_{\bar{\mu}}\|_{H^1(D)} (1 + |\omega|)^{s_1}$$

and the optimality inequality $|\bar{P}(\omega)| \leq \alpha + \beta(1 + |\omega|^p)$ bounds the support (Lemma 3.21). Consequently, local minimizers have finite support (Theorem 3.23), the sufficient condition for local optimality only has to be checked on a compact set (Theorem 3.28), and the insertion problem is coercive (Corollary 3.25). For positively k -homogeneous profiles, we prove the corresponding first-order conditions and the existence of a global minimizer represented by a finite network, with an explicit bound on the width.

For the Van der Pol oscillator, the displayed Gaussian comparison shows the trade-off between validation error, support size, and $R_{0.95}$ as β changes. The Gaussian gives the smallest H^1 error among the tested nonhomogeneous profiles, while softplus uses fewer atoms. The homogeneous ReLU^k networks obtain comparable H^1 errors with smaller supports for larger k .

For the pendulum swing-up, the ReLU^2 errors in Table 4 are larger on $S_{0.3}$ than on its complement, whereas the Gaussian errors are nearly equal for the first two training sets and larger on the complement for the other two. Adding training samples near S does not reduce the error on $S_{0.3}$ at the tested network widths. ReLU^2 gives the smallest error and the closed-loop path closest to the reference from both initial states. The softplus network in Table 5 also reaches the upright from both states, at a larger cost. In the softplus comparison with α , γ , and p fixed, every tested positive value of β reduces the rollout cost relative to $\beta = 0$, but none of those networks reaches the upright.

The experiments are two-dimensional, and the behavior in higher dimensions remains to be studied. For targets with a gradient discontinuity, different weights on samples near and away from S may also be considered. Finally, Corollary 3.27 identifies escaping directions normal to a flat part of the boundary. This suggests studying coercive anisotropic weights adapted to the geometry of ∂D .

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